

Lecture No. 1

Introduction of Indian agricultural heritage, Need and importance for studying Agricultural Heritage

Globally Important Agricultural Heritage Systems (GIAHS), as defined by the FAO (Food and Agriculture Organization of the UNO), are: "Remarkable land use systems and landscapes which are rich in globally significant biological diversity evolving from the co-adaptation of a community with its environment and its needs and aspirations for sustainable development".[1] Worldwide, specific agricultural systems and landscapes have been created, shaped and maintained by generations of farmers and herders based on diverse natural resources, using locally adapted management practices. Building on local knowledge and experience, these ingenious agri-cultural systems reflect the evolution of humankind, the diversity of its knowledge, and its profound relationship with nature. These systems have resulted not only in outstanding landscapes, maintenance and adaptation of globally significant agricultural biodiversity, indigenous knowledge systems and resilient ecosystems, but, above all, in the sustained provision of multiple goods and services, food and livelihood security for millions of local community members and indigenous peoples, well beyond their borders.

For millennia communities of farmers, herders, fishers and forest people have developed complex, diverse, and locally adapted agricultural systems. These systems have been managed with time-tested, ingenious combinations of techniques and practices that have usually led to community food security, and the conservation of natural resources and biodiversity. Agricultural heritage systems can still be found throughout the world covering about 5 million hectares, which provide a vital combination of social, cultural, ecological and economical services to humankind. These “Globally Important Agricultural Heritage Systems-GIAHS” have resulted not only in outstanding landscapes of aesthetic beauty, maintenance of globally significant agricultural biodiversity, resilient ecosystems and a valuable cultural heritage Above all these systems sustainable provide multiple goods and services, food and livelihood security for millions of poor and small farmers. The existence of numerous GIAHS around the world testifies to the inventiveness and ingenuity of people in their use and management of the finite resources, biodiversity and ecosystem dynamics, and ingenious use of physical attributes of the landscape, codified in traditional but evolving knowledge, practices and technologies. Whether recognized or not by the scientific community, these ancestral agricultural systems constitute the foundation for contemporary and future agricultural innovations and technologies. Their cultural, ecological and agricultural diversity is still evident in many parts of the world, maintained as unique systems

of agriculture. Through a remarkable process of co-evolution of Humankind and Nature, GIAHS have emerged over centuries of cultural and biological interactions and synergies, representing the accumulated experiences of rural peoples.

Agriculture Heritage in India

Our heritage is unique than any other civilization. As a citizen of India, we must feel proud about our rich cultural heritage. Agriculture in India is not of recent origin, but has a long history dating back to Neolithic age of 7500-6500 B.C. It changed the life style of early man from nomadic hunter of wild berries and roots to cultivator of land. The wisdom gained from ancient saints and practices adopted by them have been passed down through generations. The traditional farmers have developed the nature friendly farming systems and practices such as mixed farming, mixed cropping, crop rotation etc. The great epics of ancient India convey the depth of knowledge possessed by the older generations of the farmers of India.

Need and importance for studying Agricultural Heritage

Our agriculture has lot of inherited sustainable practices passed from one generation to other generation. And also agriculture in India is not an occupation, it is a way of life for many Indian populations. Hence the present day generation should be aware about our ancient and traditional agricultural systems and practices. This will enable us to build the future research strategy also.

India has made tremendous progress in agriculture and its allied fields, but the emphasis on intensive use of inputs without considering their adverse impact of long term basis has created several problems related to sustainability of agriculture. Irrational use of chemical fertilizers, insecticides and exploration of natural resources is threatening the agro eco systems. Soil is getting impoverished, water and air getting polluted and there is an increasing erosion of plant and animal genetic resources. Therefore, attention is now shifting towards sustainable form of agriculture.

The indigenous technical knowledge (ITK) provides insight into the sustainable agriculture, because these innovations have been carried on from one generation to another as a family technology. There are several examples of valuable traditional technologies in India but unfortunately these small local systems are dying out. It is imperative that we collect, document and analyze these technologies so that the scientific principle/basis behind them could be properly understood. Once this done, it will be easier for us to further refine and upgrade them by blending them with the modern scientific technology.

Objective of the course

- To increase awareness of the rich heritage of Indian agriculture which is unique than any other civilization.
- To implant a sense of pride amongst the people, particularly agricultural students as our agriculture has sustainable practices for generations.
- To collect, document and analyze the traditional agricultural technologies so as to understand the scientific principle/basis behind them.
- To refine and upgrade those technologies by blending them with the modern scientific technology.
- To stimulate scientific research based on traditional technology and build the future research strategy.

Definitions

HISTORY : Continuous record of past events

HERITAGE : Inherited values carried from one generation to other generation

AGRICULTURAL HERITAGE : Values and traditional practices adopted in ancient India which are more relevant for present day system.

History denotes the continuous record of past events, whereas heritage indicates the inherited values carried from one generation to other generation. Agricultural heritage denotes the values and traditional practices adopted in ancient India, which are more relevant for present day system.

List of Available Documents on agriculture during ancient and medieval period

1. Rigveda (c.3700 BC)
2. Atharvaveda (c. 2000 BC)
3. Ramayana (c.2000 BC)
4. Mahabharata (c.1400 BC)
5. Krishi-Parashara (c.400 BC)
6. Kautilya's Artha-sastra (c.300 BC)
7. Amarsimha's Amarkosha (c.200 BC)
8. Patanjali's Mahabhasya (c.200 BC)
9. Sangam literature (Tamils) (200 BC-100 AD)
10. Agnipurana (c.400 AD)
11. Varahamihir's Brhat Samhita (c. 500 AD)
12. Kashyapiyakrishisukti (c.800AD)
13. Surapala's Vrikshayurveda (c.1000 AD)

14. Lokopakaram by Chavundaraya (1025 AD)
15. Someshwardeva's Manasollasa (1131 AD)
16. Saranghara's Upavanaviyoda (c.1300 AD)
17. Bhavaprakasha-Nighantu (c.1500 AD)
19. Dara Shikoh's Nuskha Dar Fanni-Falahat (c.1650 AD)
20. Jati Jaichand's dairy (1658-1714 AD)
21. Anonymous Rajasthani Manuscript (1877 AD)
22. Watt's Dictionary of Economic Products of India (1889-1893 AD)

Formation of Indian – sub continent

Earth is a dynamic and constantly changing planet. Earth's crust is divided into many pieces called plates. These plates are in constant motion causing earth quakes, volcanoes etc. They move at a speed of 1.3 to 10 cm per year. Due to the movement over years several changes have occurred which has resulted in the formation of the Indian sub-continent.

Pangaea, the super-continent

About 250 millions years ago the Earth's seven continents were all grouped together into a super continent (one huge landmass) called '**Pangaea**'. This huge super continent was surrounded by one gigantic ocean called **Panthalassa**. Australia was west of its current position. North America was far in the south-east near the tropics. India was miles away from Asia.

Laurasia

About 180 million years ago the super continent Pangaea began to break up into Laurasia and Gondwanaland. **Laurasia** was made of the present day continents of North America (Greenland), Europe, Angara land (Asia) comprising Russia, Siberia and China in the north. **Gondwanaland** was made of the present day continents of South America, Africa, India, Australia, and Antarctica. At this time India was not connected to Asia. The huge ocean of Panthalassa remained.

'**The Triple Junction**' was formed because of a three-way split in the crust allowing massive lava flows in three directions which later resulted in the separation of Africa and South America. The rocks in these two continents were produced at the same time and in the same place. This tells us that South America and Africa were connected at one time. Today these two continents are separated by the Atlantic Ocean which is over 2000 miles wide.

Laurasia was still moving, and as it moved it broke up into the continents of North America, Europe and Asia (Eurasian plate). In the second stage, the Gondwana continents

separated from each other. Later South America separated from Africa. This created another narrow basin between these two continents. The eastern coast of North America separated from the Moroccan bulge of Africa. The breakup of the Gondwanaland opened up the Atlantic and the Indian Ocean.

The Indian Subcontinent moved hundreds of miles in 135 million years at a great speed (4 inches per year). The Indian plate crashed into the Eurasian plate (Asia) with such speed and force that it created the tallest mountain range on Earth, the Himalayas. The Tethys ocean (now Mediterranean sea) was being squeezed out of existence in the east of the Alpines as India approached Asia. Marine sediments excavated from the roof of the Himalayas proves the existence of Tethys ocean.

Lecture No 2 & 3

Ancient agricultural practices, Paleolithic age (old stone age), Mesolithic period, Neolithic Agricultural Revolution Chalcolithic culture (Bronze age) and Beginning of Agriculture in India: Archeological and historical facts

Development of human culture

It is supposed that man was evolved on earth about 15 lakh years ago. This man was evolved from the monkey who started to move by standing erect on his feet. Such man has been called Homo erectus (or) Java man. Later on Java man transformed into Cro-Magnon and

Cro-Magnon into modern man. The modern man is zoologically known as Homo sapiens (Homo - Continuous, Sapiens - learning habit). In the beginning such man had been spending his life wildly, but during the period 8700-7700 BC, they started to pet sheep and goat, although the first pet animal was dog, which was used for hunting.

The history of agriculture and civilization go hand in hand as the food production made it possible for primitive man to settle down in selected areas leading to formation of society and initiation of civilization. The development of civilization and agriculture had passed through several stages. Archeologist initially classified the stages as stone age, Bronze and Iron age. Subsequently the scholars spilt up the stone age into Paleolithic period (old stone age), Neolithic age (New stone age) and Mesolithic age (Middle stone age).

Each of three ages, saw distinct improvements. The man fashioned and improved tools out of stones, bones, woods etc. to help them in day-to-day life. They started growing food crops and domesticated animals like cow, sheep, goat, dog etc.

Classification of ages:

Initially archeologists classified the time period into three:

- 1) Stone age,
- 2) Bronze age
- 3) Iron age.

Stone age divided into

- a) Paleolithic (Old stone age),
- b) Mesolithic (Middle stone age)
- c) Neolithic (New stone age)

The word 'lithium' comes from a Greek word, "lithos", (Stone) Bronze age was later called as Chalcolithic period.

Paleolithic age (old stone age) (2.5 million-12,000 BC):

This period is characterized by the food gatherers and hunters. The stone age man started making stone tools and crude choppers. He learnt to control fire, which helped him to improve his way of living. At the end of this age, the modern human being (*Homo sapiens*) first appeared around 36,000 BC.

Mesolithic period (12,000 to 7,500 BC)

The transitional period between the end of the Paleolithic and beginning of the Neolithic is called Mesolithic. It began about 10000BC and ended with the rise of agriculture. This period is characterized by tiny stone implements called microliths. People lived as food gatherers and hunters. During this time, sharp and pointed tools were used for killing fast-moving animals. Humans began to store the food in storages for later consumption. The domestication of the dog was the major achievement of the Mesolithic hunter. The beginning of plant cultivation also started.

Neolithic Agricultural Revolution (7500 BC - 6500 BC)

Neolithic revolution brought a major change in the techniques of food production which gave man control over his environment and saved him from the precarious existence of mere hunting and gathering of wild berries and roots. Domesticated animals like goat, sheep, pig and cattle. For the first time, he lived in settled villages and apart from security from hunger he had leisure time to think and contemplate.

The main features of Neolithic culture in India

1. Neolithic culture denotes a stage in economic and technological development in India
2. Use of polished stone axes for cleaning the bushes
3. Hand made pottery for storing food grains
4. Invented textile, weaving and basketry
5. Cultivation of rice, banana, sugarcane and yams in eastern parts of India
6. Cultivation of millets and pulses in south India
7. Discovery of silk
8. Domesticated horse and ass
9. Distinctive feature – Development of house built of locally available materials

Chalcolithic culture (Bronze age) (3000-1700 BC):

The term Chalcolithic is applied to communities using stone implements along with copper and bronze. In more advanced communities, the proportion of copper and bronze implements is higher than that of stones. The chalcolithic revolution began in Mesopotamia in the fourth millennium B.C. from this area it spread to Egypt, and Indus valley. Copper was used by Egyptians and Sumerians.

The significant features are

1. Invention of plough
2. Agriculture shifted from hilly area to lower river valley
3. Flood water was stored for irrigation and canals were dug
4. Irrigated farming started in this period
5. Sowing of seed by dibbling with a pointed stick
6. Salinity problem and water logging were noticed due to canal irrigation.

Beginning of Agriculture in India: Archaeological and historical facts

12000 to 9500 years ago

- ❖ Hunters and food-gathers stage existed.
- ❖ Stone implements (Microliths) were seen throughout the Indian subcontinent.
- ❖ Domestication of dog occurred in Iraq.
- ❖ Earliest agriculture was by vegetative propagation (e.g. bananas, sugarcane, yam, sago, palms, and ginger).

9500 to 7500 years ago

- ❑ Wild ancestors of wheat and barley, goat, sheep, pig, and cattle were found.

7500 to 5000 years ago

- ❖ Significant features were invention of plough, irrigated farming, use of wheel, and metallurgy and in Egypt, seed dibbling.

5000 to 4000 years ago

- ❖ Harappan culture is characterized by cultivation of wheat, barley and cotton; plough agriculture and bullocks for drought.
- ❖ Wheeled carts were commonly used in the Indus valley.
- ❖ Harappans not only grew cotton but also devised methods for ginning / spinning / weaving.

4000 to 2000 years ago

- ❖ In North Arcot, bone / stone tools were found.
- ❖ In Nevasa (Maharashtra), copper and polished stone axes were used. First evidence of the presence of silk was found at this location.
- ❖ At Navdatoli on Narmada river (Narmar, Madhya Pradesh), sickles set with stone teeth were used for cutting crop stalks. Crops grown were wheat, linseed, lentil, urd (black gram), mung bean, and khesari.
- ❖ In Eastern India, rice, bananas, and sugarcane were cultivated.

2000-1500 years ago

- ❖ Tank irrigation was developed and practiced widely.
- ❖ Greek and Romans had trade with South India; pepper, cloth, and sandal wood were imported by Romans.
- ❖ Chola King Karikala (190 AD) defeated Cheras and Pandyas, invaded Srilanka, captured 12000 men and used them as slaves to construct an embankment along the Cauvery, 160km long, to protect land from floods. He has built numerous irrigation tanks and promoted agriculture by clearing forests.

1500-1000 years ago

The Kanauj Empire of Harshavardhana (606-647 AD)

- ❖ Cereals such as wheat, rice and millets, and fruits were extensively grown. A 60-day variety and fragrant varieties of rice are mentioned.
- ❖ Ginger, mustard, melons, pumpkin, onion, and garlic are also mentioned.
- ❖ Persian wheel was used in Thanesar (Haryana).

The kingdoms of South India

- ❖ The kingdoms were of the Chalukyas (Badami), Rashtrakutas (Latur), Pallavas (Kanchi), Pandyas, Hoysals (Helebid), and Kakatiyas (Warangal).
- ❖ Cholas ushered in a glorious phase in South Indian in the 10th century AD.
- ❖ New irrigation systems for agriculture were developed- chain tanks in Andhra in the 9th century; and 6.4km Kaveripak bund.
- ❖ Cholas maintained links with China, Myanmar, and Campodia.
- ❖ The tank supervision committee (Eri-variya) looked after the maintenance of a village and regulated the water supply.

1000-700 years ago

- ❖ Arab conquest of Sind was during 711-712 AD; Md bin Qaism defeated Dahir, the Hindu king of Sind. Arabs were experts in gardening.
- ❖ 1290- 1320AD (Reign of Khiljis): Alauddin Khilji destroyed the agricultural prosperity of a major part of India. He believed in keeping the farmers poor.

With the introduction of agriculture,: villages and cities became possible as people did not have to travel in search of food. (*Civilisation* comes from the Latin word "*civitas*" meaning city.)

This sedentary way of life formed the basis for modern civilization.

Technological advancements in ancient periods

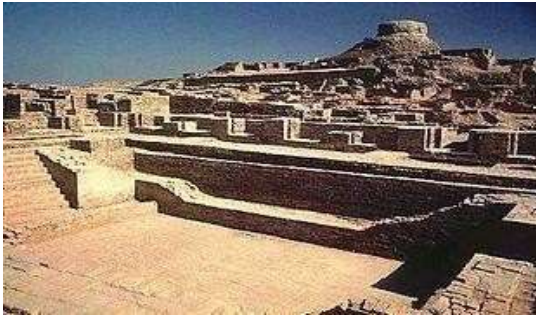
Technology	Place of origin	Year
Seed drills	Mesopotamia	5500 years ago
Irrigation systems	Egypt, Mesopotamia	5000 years ago
Iron plough (more effective tool)	China	2600 years ago
Mould board plough	China	2100 years ago
Crop rotation, Horse drawn plough	Europe	AD 1000
Water wheels (for grinding corn or sawing wood)	England	AD 1066

Lecture No 4

Indus civilization, Vedic civilization and Relevance of heritage to present day of agriculture

Indus Civilization

few



In the year 1922, archaeologists dug up a places in the Indus valley and carried out excavations at Mohenjodara (meaning a mound of dead) in Sind (in Pakistan) and at Harappa on the river Ravi in Punjab. They

found traces of a very ancient civilization, which flourished more than five thousand years ago.

They observed that the

people utilized the pots, utensils and ornaments. These cities were built along the river Indus and hence this civilization is known as Indus valley civilization. It is



also known as **Harappan culture** and occupied the areas stretching from Delhi to Gujarat.



- ❖ It confined to the valley of the river Indus.
- ❖ It was a highly developed urban one and two of its towns, Mohenjo-daro and Harappa, represent the important settlements.
- ❖ Subsequent archaeological excavations established that this civilization were not restricted to the Indus valley but spread to a wide area in northwestern and western India.
- ❖ It is now better known as the Harappan civilization.
- ❖ Mohenjo-daro and Harappa are now in Pakistan
- ❖ Principal sites in India include Ropar in Punjab, Lothal in Gujarat and Kalibangan in Rajasthan.
- ❖ The Indus Valley Civilization stretched across the whole of Sindh, Baluchistan, Punjab, Northern Rajasthan, Kathiawar and Gujarat.

During this period the people identified the importance of ploughing for the proper sowing

of crop (i.e) soil has to be stirred and seed has to be covered. Ox-drawn wheel cart was used for transport. The people cultivated wheat, barley, gram, peas, sesamum and rape. They also cultivated cotton and also devised methods of ginning, spinning and weaving. Animal husbandry was also given more importance during this period. They domesticated buffalo, cattle, camel, horse, elephant, ass and birds. They utilized them in agriculture and also for transport.

The most remarkable discovery in Harappa is the Great Granary used for storing food grain. These granaries, each 50x20 feet overall, are arranged symmetrically in two rows of six in each row with central passage and 23 feet wide. From the size of the granary it can be concluded that the peasants paid their dues to the Government in kind, used the kinds in granary for payments to employees. The artisans, carpenters and others received their wages in kind from the farmers.

Developments in agriculture and trade

- ❖ The economy of Indus valley civilization appears to have depended significantly on trade, which was facilitated by major advances in transport technology.
- ❖ These advances included bullock-driven carts that are identical to those seen throughout South Asia today, as well as boats.
- ❖ Agriculture have been highly productive; as it was capable of generating surpluses sufficient to support tens of thousands of urban residents who were not primarily engaged in agriculture.
- ❖ It relied on the considerable technological achievements of the pre-Harappan culture, including the plough.
- ❖ People built their lives around the monsoon, a weather pattern in which the bulk of a year's rainfall occurs in a four-month period.
- ❖ During this period the people identified the importance of ploughing for the proper sowing of crop (i.e) soil has to be stirred and seed has to be covered.

The Vedic civilization



The word “Veda” is derived from “Vid” which means “Knowledge” Veda is the only literary source from which we know about the Aryans in India. Aryans were more prevalent during Vedic time which extends from Eastern Afghanistan, Kashmir, Punjab and Parts of Sind and Rajasthan. The land of Aryans was called land of seven rivers i.e., (Satlaj, Beas, Ravi, Chennab,

Jhelum, Indus and Saraswathi). The Rig-veda was the oldest book of Aryans.

Status of Agriculture in the Vedic period

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- ❖ Early vedic Aryans were primarily pastoral and settled in Indus valley.
- ❖ They cut jungles, built their villages, grazed their cattle in jungles and planted barley in the land close to their habitation.
- ❖ Vedic Aryans were adopted cows, horses; buffalo was a new animal which they called gauri or govala which appears to be an extension of the word gau (cow).
- ❖ Indus valley is the land of seven rivers was called 'Saptasindhavah'.
- ❖ The seven rivers included the five rivers of the Punjab (meaning land of five rivers) viz., the Sutudri (Sutlej), the Vipas, (Beas), the Parushini (Ravi), Jhelum and the Askini (Chenab) and the remaining two includes the Indus and Saraswathi.

Environment (Rigveda):

- ❖ There are six seasons in a year; namely Grishma (May –June), Varsha (July-August), Hemant (September-October), Sharad (November-December), Shishir (January-February) and Vasant (March-April).
- ❖ The beginning of the rainy season (obviously in Pakistan North-West India) is after 21 June when the sun starts 'moving south'.
- ❖ There were dams on the seven rivers.

Constructing dams on rivers must have meant cutting off water to Vedic people to irrigate lands and to provide water to people and animals after the rains for increasing the food production

Pastoralism

The Vedic Aryans were primarily pastoral. When they settled in the Punjab, they cut the jungles and built their villages. They grazed the animals in jungles and cultivated barley near the houses to protect from wild animals.

Vedic people realized the importance of off-season ploughing and they started ploughing as and when the rain was received. The first ploughing of the season was inaugurated amidst much ritual. The plough used was large and heavy. Bullocks and ox were used for ploughing. With regard to irrigation, channels were dug from the rivers. Wells were in use for supply of drinking water and irrigation called kucha wells, which were just holes dug in the ground. Even now such wells are in use in the river rain areas of northern India.

Crops cultivated in Vedic period

In early Vedic period there is no mention of rice and cotton though they were cultivated in Harappa period. In the later Vedic period (1000 - 600 BC) agricultural implements were improved and iron ploughshare also improved. The people possessed the knowledge of fertility of land, selection of seed, seed treatment, harvesting, manuring and rotation of crops. Barley sesame and sugarcane were the main crops. Cucumber and bottle gourd were also mentioned in Vedic period,

Aryans were accustomed to barley diet. Barley is good for men, cattle and horses. Barley is used in Hindu rituals even today. For cloths, wool and cotton were used.

The agriculture implements mentioned in vedic literature include the plough (langala - a lase pointed type having smooth handle, Sira - a large and heavy plough).Sickle was used for harvesting and sieves were used for cleaning.

Civilizations in other parts of World

LEMURIA CIVILIZATION

Lemuria was originally the name given to a vast hypothetical sunken continent or a land-bridge or landmass stretching from Ceylon to Madagascar all the way to the central Pacific Ocean across the Indian Ocean and Indonesia. Ancient Lemuria-map of India in 30,000 B.C. The lemurs derive their name from that of the Lemurs (or "Ancestors"). Man descends from the apes. Hence, the name of Lemuria can be interpreted as "Land Ancestral" or "Land of the Ancestors". The ancient land tying India and Australia together that sank incrementally over time, is referred to as 'Lemuria'. The Tamil bark writings in Southern India tell of the gigantic Southern part of India, which used to connect to Australia cataclysmically sinking incrementally over a long period of time. This was ancient Lemuria or Kumari Kandam. The great flood would have sunk Lemuria or Kumari Kandam before the Ramayana period (10,000 BC).

Differentiate Between Indus Valley Civilization and Vedic Civilization

Sr No	Indus Valley Civilization	Vedic Civilization
1	It is a civilization developed by the Dravidian people	It is a civilization developed by the Aryans
2	It was an urban civilization	It was mainly the rural civilization,
3	Reflected in city life	Centered round the village
4	The people did not know the use of iron	
5	They used the materials like Stone, Copper and Bronze in making weapons	The people used iron weapons extensively
6	The people did not know the use of horse in war	The people had the cavalry. They introduced horse in India
7	Mother cult as well as the cult of Siva were prominent in the religious belief	People believed in one God
8	Bull were of high esteem	Cows were the high esteem animals
9	No social division was found among the people	Social divisions were conspicuous in the society

Lecture 5,6 and 7

Status of farmers in the society during Indus, Vedic, Buddhist, Mauryan, Gupta and Sangam periods

Status of farmers in the society during following periods

Indus / Harappan period:

The Indus Civilization had the first farming cultures in South Asia, which emerged in the hills of what is now called Baluchistan, to the west of the Indus Valley.

The farmers took part in the so-called Neolithic Revolution, which took place in the Fertile Crescent around 9000 to 6000 BCE.

The **Neolithic Revolution, Neolithic Demographic Transition, Agricultural Revolution, or First Agricultural Revolution** was the wide-scale transition of many human cultures during the Neolithic period from a lifestyle of hunting and gathering to one of agriculture and settlement. These settled communities permitted humans to observe and experiment with plants to learn how they grew and developed.^[2] This new knowledge led to the domestication of plants.

□ These early farmers domesticated wheat and a variety of animals, including cattle.

In the "Era" terminology, the Neolithic is known as the "Early Food Producing Era".

Early Harappan

- ❖ The development of these farming communities ultimately led to the formation of larger settlements.
- ❖ Indus valley civilization was composite product of different races (Africans, Europeans and Asians) who lived and worked together in a particular environment.
- ❖ Mohenjo daro had easy land and water communication; it was the meeting ground of people for different parts of Asia.
- ❖ Farmers had, by this time, domesticated numerous crops, including peas, sesame seeds, dates and cotton, as well as a wide range of domestic animals, including the water buffalo.

Late Harappan

- ❖ By 2500 BCE, the Early Harappan communities had been turned into urban centers.
- ❖ Thus far, six such urban centers have been discovered, including: Harappa, Mohenjo Daro and Dholavira in Pakistan, along with Gonior, Kalibangan and Mangalore in India.
- ❖ In total, over 1052 cities and settlements have been found, mainly in the general region of the Ghaggar-Hakra River and its tributaries.
- ❖ By 2500 BCE, irrigation had transformed the region.

Ox-drawn wheel cart was used for transport.

- ❖ Farmers were an important part of Harappan civilization.
- ❖ Perhaps the main work of the inhabitants of these rural settlements was agriculture.
- ❖ The bullocks are very important in this work so this importance is shown on the seals and toys.
- ❖ Animal husbandry was also given more importance during this period. They domesticated buffalo, cattle, camel, horse, elephant, ass and birds. They utilized them in agriculture and also for transport.



- ❖ The most remarkable discovery in Harappa is the Great Granary used for storing food grain.
- ❖ These granaries, each 50x20 feet overall, are arranged symmetrically in two rows of six in each row with central passage and 23 feet wide.
- ❖ The "granary" of Harappa is a brick structure that was built on a massive brick foundation and has three sleeper walls with air space between them.
- ❖ Large site bases, storage bins and the structures identified as possible granaries in the living complexes give us a clear idea of the strong agricultural base in the economy also.

- ❖ From the size of the granary it can be concluded that farmers of this as Indus valley civilization produced sufficient food grains not only for their consumption but also for the town peoples as there was **barter system** that is the peasants paid their dues to the Government in kind, used the kinds in granary for payments to employees.
- ❖ The artisans, carpenters and others received their wages in kind from the farmers.

So this shows that entire crop is not consumed immediately. Some quantity of grain is conserved so as to last until the next harvest and for use during lean seasons, famine and drought or crop failure. Some portion is kept aside as seed for the next sowing. So the storage facility also helps us the evidence of type of grains were in use at that such time and also helps in understanding the type of grains was consumed during that period.



Vedic period (1500-500):

In early Vedic period there was no mention of rice and cotton though there were cultivated in Harappa period. In the later Vedic period (1000 – 600 BC) agricultural implements were improved. Iron ploughshare was also improved. The people possessed the knowledge of fertility of land, selection of seed, seed treatment, harvesting manuring and rotation of crops. Barely sesame and sugarcane were the main crops. Cucumber and bottle ground were also mentioned in Vedic period, Aryans were accustomed to barley diet. Barley is good for men cattle and horses. Barley is used in Hindu rituals even at present. For cloths, wool and cotton were used.

The agriculture implements mentioned in Vedic literature include the plough (langala, - a lase pointed type having smooth handle, Sira – a large and heavy plough). Sickle was used harvesting sieves were used for cleaning.

The most important people of the Vedic period are Bramhanas. There are four Vedic periods viz., Rig, Sama, Yajur, and Atharvana Vedas. In Rig Vedas period, the farmers occupied more number in the society.

During this period, the superior people are called as Vaishnavas, the next position was Shathriyas and the least position occupied was Suthriyas. The Suthriyas are the farmers they cultivated the land and produced agricultural products under the land lord. The farmers status was more in Atharva Vedic period. They cultivated the crops based on the advice of the saints.

- ❖ **The early vedic** period was from 1500 - 1000 BCE and consists primarily of the samhita of the Rig veda.
- ❖ During this period the society was at a primitive stage.
- ❖ People were agro-pastoral and gatherers and a man's wealth was measured by the number of **cows he had**.
- ❖ There were **no fixed varnas** and people could practice any profession they like.

Early vedic people had a tribal life and then gradually evolved to territorial state.

- Occupation of individuals was the basis of classification of a society in the Rig Vedic period.
- It was divided into four *varnas*, namely
 - *Brahmanas* (teacher and priests);
 - *Kshatriya* (rulers and administrators);
 - *Vaisya* (farmers, merchants, and bankers); and
 - *Sudra* (artisan and laborers).
- There was complete freedom and mobility for the adoption of a profession.
- Trades and occupations did not assume a hereditary character in the society (till now).
- Agriculture, cattle rearing, and trade and commerce were the main economic activity of the Rig Vedic people.
- People had domestic animals like cows, sheep, goats, asses, dogs, buffalos etc.
- Oxen were used for ploughing and drawing carts and horses for drawing the chariots.
- The plough was drawn by the oxen at times in a team of six, eight, or even twelve.
- The grains were harvested with sickles.

- Manure was used for high yield; irrigation was also practiced.
- Excess of rains and drought is mentioned as damaging the crops.
- The grains are collectively called '*Yava*' and '*Dhanya*.'
- The practices of exchange of goods (Barter Economy) were in trend.

LATTER VEDIC PERIOD

- During the **latter Vedic period**, *Varnas* came to be birth-based rather than profession-based (as were in Rig Vedic period).
- Development of new professions gave rise to '*jatis*.' But the *jati* system was not yet as rigid as it became during the period of the *sutras*.
- Atharvanaveda dealt with economic aspect. It described many prayers to bring economic prosperity for the success of farmers, shepherds, merchants, and so on.
- Atharvanaveda explained prayers for ploughing, sowing, rains, and an increase in cattle, wealth, and exorcism against beasts, wild animals, and robbers.
- The plough was known as ***Sira*** and the furrow ***Sita***.
- Cow dung was used as manure.
- It is mentioned that six, eight, and sometimes even twenty-four oxen were used to pull a plough.
- Many types of grains were grown, such as rice, barley, beans, and sesame. Their seasons are also mentioned as barley sown in winter, ripened in summer; rice sown in the rains, reaped in autumn and so on.
- The Atharvanaveda discussed that drought and excess rains threatened agriculture.
- The Atharvanaveda mentioned that hymens to worship the cow and the death penalty were prescribed for cow killing.

Buddhist period:

A food producing economy emerged with the practice of agriculture on a wide scale by using iron implements. There was pleasant proprietorship in rural areas and there were no land lords. But a land owner could not sell for mortgage his land without permission of the village councils. The village residents together undertook task such as laying irrigation channels, buildings, rest houses etc. the women extended their full co-operation in their works (public utility). The whole of each village was self-sufficient, life was simple.

Mauryan period:

In this period the economy was agrarian, majority of population were agriculturists. People were also engaged in animal husbandry and cattle rearing which meant additional income to peasants and the state. Gaha pathi were the term used for head of rich land owning family.

- ❖ Society and Culture during the Mauryas were well classified and organized; work of every class was decided accordingly
- ❖ Largely, the population was agriculturists and lived in villages. The state helped people to bring new areas under cultivation by cleaning the forest. But certain types of forests were protected by law.
- ❖ A number of crops like rice, coarse grains (*kodrava*), sesame, pepper, and saffron, pulses, wheat, linseed, mustard, vegetable and fruits of various kinds and sugarcane were grown.
- ❖ The state also owned agricultural farms, cattle farms, dairy farms, etc.
- ❖ Water reservoirs and dams were built by the state for irrigation. Steps were taken to distribute and measure this water for irrigation.
- ❖ The Mauryan enforced the rules and regulations in respect of agriculture, industry, commerce, animal husbandry, etc.
- ❖ Pushyagupta (Chandragupta's governors) was responsible for building a dam on *Sudarshana* Lake near Girnar in Kathiawad.
- ❖ They had foreign trade with the western countries. The main items of trade were indigo, various medicinal substances, cotton, and silk. The foreign trade was carried on by the land as well as by the sea.

Gupta period:

- ❖ This period is called golden age of India. Provides information on agriculture besides other sciences. Deals with selection of land, manuring, cultivate, seed collection, sowing, planting & grafting.
- ❖ The cultivators were called by various terms called Krishihala or Kinars. They had low social and economic life.
- ❖ Agriculture during the Gupta Empire was bountiful.
- ❖ The rivers of Ganges, Brahmaputra, Narmada, Godavari, Krishna, and Kaveri provided the source of irrigation.
- ❖ With Gupta administration constructing numerous irrigation infrastructures such as aqueducts, dams, and canals, large areas of the Empire became very fertile for farmers to till.
- ❖ Various crops, important to the diet of the Indians, were grown. This includes rice, wheat, barley, peas, and lentils.
- ❖ Spices in the southern region of the Empire.
- ❖ Landownership and sharecropping system was in existence. Land transfers must be made in presence of the village council. The state also owned lands called Rajayavastu, which were usually given to landless farmers.

Sangam period:

During Sangam literature, agriculture was the main occupation and hence the position of the farmers in the society was also high during this period. Agriculture Sangam was developed in Madurai. The farmers are called uzhavar (plough man) and also they are called as Kalmar. The land owners called superior vellars and the farmers who plough the land are called as inferior vellars. The farmers' status was mainly determined by the holding of land and animal population. Agriculture was considered a necessity for life, and hence was treated as the foremost among all occupations.

PRESENT STATUS OF MODERN AGRICULTURE

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Agriculture plays a crucial role in the development of the Indian economy. Modern agriculture has made impressive contribution in increasing food grain production in the country. The country could achieve self-sufficiency in food grain production by using modern methods of agriculture using better quality of seed, proper irrigation and adequate supply of plant nutrients by using chemical fertilizers and control of pests and diseases in crop plants by using pesticides. It has also involved modern cultivation practices using tractors, combine harvesters and tube wells for irrigation. Rapid growth in food grain production from using seeds of high yielding variety is termed as Green Revolution. Protection of air, water, soil and human health while producing bumper crops should be the prime concern of modern agriculture.

- ☐ Today we are second largest producer of wheat, rice, fruits, vegetables, and fresh water aquaculture; and largest exporter of spices and cashew. The late sixties and seventies were the years of Green Revolution. During Yellow Revolution oilseeds production reached up to 24.4 million tonnes.
- ☐ Per capita availability of food grains went up to 528.77 g per day in 1996-97 when compared to 395 g in early fifties. Fertilizers consumption has also increased and India has become fourth in the world after USA, USSR and China. Pulse crops are grown on the largest Indian area in the world and India is the first to evolve a cotton hybrid.
- ☐ Cropping pattern is changing and commercial crops and non-traditional (moong, soya bean, summer groundnut, sunflower etc.) are gradually growing more importance in line with domestic demands and export requirements. Short duration varieties have been introduced to use the residual moisture available from post-kharif and post-rabi cultivation.
- ☐ The 3 main crop seasons are – kharif, rabi and jayad. Major crops are rice, jowar, bajra, maize, cotton, sesame, soya bean and groundnut. Major rabi crops are wheat, jowar, barley, gram, linseed, rapeseed and mustard. Rice, maize and groundnlit are grown in the summer also.

Agricultural Development Projects (ADPs) with the assistance of World Bank are being implemented to (i) enhance long-time sustainability in agriculture, and (ii) create infrastructure in rural area.

Lecture No 8 & 9

Plant production through indigenous traditional knowledge

Beginnings of formal agricultural research and extension systems in the country aimed at exploiting indigenous or traditional technologies.

Even within traditional systems, gaps existed between 'good' and 'bad' farmers and practices.

Both research and extension systems sought to exploit the 'good' practices.

These technologies have five major characteristics.

- a) First, they have low capital intensity, whatever capital is built up (like bunds, terraces, wells) are inadequate. They focus on improving quality of management (operations) and materials (seeds, land, tools, etc.). In capital-starved surplus labour, traditional agricultural system, this meshes ideally with relative factor scarcities.
- b) These are usually environment and ecology- friendly, these are sustainable. There are rice fields in the Indo-Gangetic Plains which have been in cultivation for very long time and produced stable output for centuries.
- c) These are generally very location and site-specific and have limited adaptability. That is why these are rarely noticed by outsiders in a macro-oriented and constrained information system.
- d) These diffuse over small, homogenous zones or sub-zones, mainly by farmer to farmer interaction. Since farmers as a group have a very low propensity for mobility, the diffusion area is further constrained. It is left to the few adventurous and pilgrimage-prone farmers, to collect varieties and ideas during their long journeys.
- e) Indigenous innovations generally generate only small increments in output. Usually, they relate to one or two practices and not the whole package. This micro orientation implies that the gains arise only from small interaction effects. Some of the related terms are:

Indigenous Knowledge (IK):

Is the participants' knowledge of their temporal and social space. Indigenous knowledge as such refers not only to knowledge of indigenous peoples, but to that of any other defined community.

or

It is the participant's knowledge of their temporal and social space. It is knowledge of indigenous peoples or defined community.

Indigenous knowledge system (IKS): Delineates a cognitive structure in which theories and of nature and culture are conceptualized. Thus includes definitions, classifications and concepts of the physical, natural, social, economic and intentional environments.

Indigenous Traditional Knowledge (ITK): Is specifically concerned with actual application of thinking of the local people in various operations of agriculture and allied areas.

Indigenous Traditional Knowledge (ITK): Is the actual knowledge of a given population that reflect experiences based on tradition.

Indigenous Technical/Traditional knowledge: in a community, local and rural in origin. The source of ITK our ancestors who learned techniques from their past experiences and experiments. These vary from place to place and knowledge spreads through folk songs, stories & scriptures etc.

Indigenous Technical Knowledge (ITK) is defined as the sum total of knowledge and practices which are based on people's accumulated experience in dealing with situation and problems in various aspects of life and such knowledge and practices are special to a particular culture.

Belief:

Change in behaviour of insects, animals and vegetation indicating a forthcoming event without any scientific rational but could be true in happening.

Innovation:

Outside the area of ITK, but scientifically based development of practices using the locally available resources to solve specific problems.

ITKs are based on experience, often tested over a long period of use, adapted to local culture and environment, dynamic and changing and lays emphasis on minimizing the risks rather than maximizing the profits.

The Indigenous Traditional Knowledge (ITK) system has been developed by the based on their experiences and continuous improvement through informal experimentation over centuries. India has one of the largest collections of ancient manuscript in the world, which includes 14 Sastras, 4 Vedas, 4 Upvedas and 6 branches of Vedangas. The advent of the concept of sustainability in Indian agricultural scenario has invoked interest on indigenous technical knowledge that has the element use of natural products to solve problems pertaining to agriculture and allied activities. ITKs are based on experiences often tested over a long period of use, adapted to local culture and environment, dynamic and changing and lays emphasis on minimizing the risks rather than maximizing the profits. ITK covers a wide range of subjects such as crop production, livestock rearing, natural resource management, food preparation, health care and many other related topics. Various aspects of agriculture and allied activities have been included this project such as on soil, water and nutrient management: crop cultivation plant protection, farm equipment, farm power, post-harvest preservation and management, pasture and fodder

management agro-forestry: bio-diversity conservation and exploitation, animal rearing and health care, animal products preservation and management; fisheries and fish preservation and ethnic foods are homestead management encompass ITKs

The differences between traditional knowledge system and scientific system

S. N.	Indigenous Traditional Knowledge system	Scientific System (Modern Agriculture)
1	All parts of the natural world are regarded in living forms, all life forms as interdependent.	Human life is generally regarded as superior, with a moral right to control other life forms
2	Knowledge is transmitted largely through oral media.	Knowledge is transmitted largely through the written word
3	Knowledge is developed and acquired through observation and practical experience	Knowledge is generally learned in a situation, which is remote from its applied context
4	Knowledge is holistic, intuitive, qualitative and practical	Knowledge is essentially reductionist, quantitative, analytical and theoretical.
5	Knowledge is generated by resource users in a diachronic (long term) time scale.	Knowledge is generated largely by specialist researchers on a synchronic (short term) time scale
6	The nature and status of particular knowledge is influenced by socio cultural factors such as spiritual beliefs, and is communally held	The nature and status of particular knowledge is influenced or peer review, and is held by individual specialists
7	Explanations behind perceived phenomena are often spiritually based on subjective	Explanation behind perceived phenomena are essentially rational and objective
8	Knowledge is used to make suitable decisions under variable conditions	Knowledge is used to put forward hypothesis and to verify underlying laws and constants

Characteristics of Indigenous Traditional Knowledge (ITK)

1. It is not static but dynamic
2. The exogenous knowledge and endogenous creativity brings change in to ITK
3. It is intuitive in its mode of thinking
4. It is mainly qualitative in nature
5. For its study needs a holistic approach.
6. ITK, if properly tapped, can provide valuable insights into resources, processes, possibilities and problems in particular area.
7. ITK is recorded and transferred through oral tradition.
8. It is learned through observation and hands-on experience
9. It forms an information base for variety
10. It reflects local tradition.

The following appropriate indigenous agriculture technologies were developed

Summer ploughing conserves moisture, eradicates weeds, consolidates soil erosion and minimizes the number of ploughings at the time of sowing.

Due to cow dung coating for cotton seeds, the easy dibbling of seeds to remove fuzz, good germination, no cost and pest-reduction were the advantages

Soaking sorghum seeds in cow urine before sowing increase the drought tolerance and the seeds had germination with minimum rain and it was considered as no cost practice.

Soaking Bengal gram in water before sowing because it considered as no cost and withstanding water stress.

Cotton seeds treated with red soils facilitate easy dibbling of seeds and it favours good germination.

Cattle penning practices improve the soil fertility owing to organic matter.

Sorghum mixed with lab-lab given additional yield owing to mixed cropping and it enhances fixation by leguminous lab-lab.

Use of cow dung cake as burrow fumigant is economical in controlling rats.

Raising of castor as a border crop in cotton field is used as trap crop for the cotton pest and it also additional income.

Easy removal of pest, easy separation of kernels, longer shelf life and higher economics are advantage of coating red gram with red soil.

Mixing green gram with ash was one of the post-harvest indigenous practice and it had certain advantages like pest reduction and cheaper method.

Crop wise Indigenous Traditional Knowledge

RICE

1. Treatment of paddy seeds in diluted bio gas slurry for 12 hours increases resistance of seedlings pests and diseases
- 2 About 30 kg of tamarind seeds are applied for an acre of paddy field 1 day after transplanting to boost up the crop growth and yield.
3. Soaking the paddy seeds in diluted cow's urine before sowing, considerably reduces the incidence of leaf spot and rice blast.
4. Presoaking of paddy seeds in milk increases its resistance against "tungro" and 'stunt' virus.
5. For control of red leaf spot disease in paddy, the seeds are soaked in 'Pudina' leaf extract (*Medicago sativa*) for 24 hours.
6. T" shaped bamboo stands are placed in many places in the paddy fields so that birds can sit on them and feed on the larvae and adults of rice pests.

7. Dhainchana (*Sesbania spp.*) seeds are sown on paddy main fields when paddy nursery is raised and is grow up dhaincha is ploughed in-situ during field preparation.
- 8 Plough the main field for four to six times for belter yield.
9. Good harvest can be obtained from the crop transplanted during Aavani i.e. Aug.- Sept.
- 10 The crop transplanted during October-November will give reduced yield.
11. The rice crop will establish better if it is transplanted along the wind direction.
12. Planting the 'samba (Aug), crop thickly and 'navarai (Feb.) thinly.
13. Practice sheep penning during summer to get more yield.
14. Practice sheep penning for the first season and green leaf manure for the second season for better yield.
15. Apply 100 kg of pig manure for one acre of rice at 10 days after planting to get higher yield.
16. Apply the neem seeds (@ 40 kg/acre as basal to get more yield as compared to the equal quantity of neem cake.
17. Irrigate the fields, allow the weed seeds to germinate and then plough the fields to incorporate the weeds into the soil before sowing or transplanting of rice crop to control weed growth.
18. Cultivation of sunhemp or daincha helps to control the nut grass (*Cyperus rotundus*) weed.
19. Neem (*Azadirachta indica*) oil cake extract is sprayed to control thrips in rice
20. Neem oil is mixed with water @ 30 ml./lit. and sprayed to control stem borer in rice.
21. To control the ear head bugs, 10 kg of cow dung ash is mixed with 2 kg. of lime powder and 1 kg of powdered tobacco waste and dusted on the rice crop during morning hours.
22. Applying neem cake before last plough to control root rot and nematode problem.
23. A mixture of 5 kg. of common salt and 15 kg. of sand is applied for 1 acre to control brown spot disease.
24. Soaking the paddy seeds in 20% mint leaves solution before sowing will control the brown leaf spot.
25. Use large mud pots called 'Kudhir' as high as six feet for storing paddy grains for longer periods.

MAIZE

1. If the sheaths on maize cobs are not removed, they can be stored for more than three months.
2. Maize seeds are soaked in cow urine for 12 hours before sowing.
3. Before sowing, maize seeds are soaked in warm water for 3-6 hours and shade dried to induce better germination and to control shoot borer.
4. Maize is sown after tomato in the same ridges and furrows to reduce the cost of land preparation.

5. When a sample of maize grains is chewed, metallic sound indicates their optimal dryness.
6. Dried maize stalks are stacked as heap on stone slabs and covered with paddy straw. This cob was stored for more than a year and used as cattle feed

Pearlmillet

1. For bird searing, the carcass of a crow is hung on a long pole in the center of the pearl millet field.
- 2 Pearl millet seeds are mixed with the leaves of noichi (*Vitex negundo*) and then stored.
3. Pearl millet is stored by building a structure called Pattarai before threshing.
4. Pearl millet ear heads are circularly to 1 foot height .
5. When pearl millet seeds are in salt solution (1 kg, of common salt in 10 lit. of water) and the floaters and disease affected seeds are removed.
6. Pearl millet seeds are presoaked in water overnight before sowing for early germination.
7. Pearl millet nursery is grown on raised beds for better seedling growth.
8. Before transplanting, seedling tips are clipped to remove the eggs of pests and white flies and to control tip blight disease.
- 9 Dusting Chula ash in pearl millet fields to control green leaf hoppers sitting on inner side of leaves
10. Grow cowpea as intercrop to control leaf miner in pearl millet.
11. Storing pearl millet seeds by mixing with ash.

SORGHUM

1. Sorghum car heads are sun dried for two days and stored without seed separation by building a storage structure called Pattarai'. This structure is opened and threshed when needed.
2. Soak the sorghum seeds in common salt solution before sowing to secure good germination under adverse conditions
3. Soak the sorghum seeds in cow urine for half-an-hour and sun drying them before sowing to control head smut and to induce drought tolerance.
4. To ensure quick germination of sorghum seeds and to avoid shoot fly attack, enough water is boiled and kept in an open place throughout the night for cooling. In the next day morning prior to sowing, the sorghum seeds are immersed in cold water for some time and sown in the field, which produces better seedlings.
5. Country plough is run at the early stage of sorghum crop to ensure optimum plant population.
6. Sow sorghum during the Tamil months Vaikasi - Aani (May-June) to avoid shoot fly and stem borer
7. Sow cowpea as an intercrop in sorghum to minimize stem borer attack due to its repellent smell.

8. Sow lab-lab as an intercrop to reduce stem borer damage in sorghum.
9. Pouring neem cake extract drop by drop on the sorghum shoot to control shoot borer
10. Dusting ash on the infected leaves of sorghum to prevent the pest incidence.
11. Dusting ash at milking stage to control ear head bugs
12. Growing coriander as a mixed crop in sorghum to control the parasitic weed viz. witch weed (*Striga lutea*).
13. A black cloth is tied to a long pole and fixed in the centre of the field to scare away the crows.
14. Mixing sorghum seeds with ash to prevent storage pests

FINGERMILLET

1. Long duration varieties are adopted in dry lands to avoid the rainy days coinciding with the harvesting stage.
2. Finger millet seeds are treated with cow urine at 1:10 ratio to enhance germination
3. When a sample of dried finger millet grain is chewed, metallic sound indicates its dryness.

RED GRAM

1. Spray the decoction of tobacco waste to control sucking pests and caterpillars.
2. Red gram seeds are mixed with red earth slurry, dried and stored to avoid storage pests.
3. Castor seeds are fried, powdered and mixed with red gram seeds to reduce pest attack during storage.
4. Dry the red gram seeds well and store them in gunny bags after placing dried leaves of 'Naithulasi' (*Ocimum canum*) inside them to prevent pod borer attack.
5. Putting the pods of dried chillies in the red gram container to control bruchids (beetle) attack.

BLACK GRAM

1. When a wooden plank is moved with pressing over the drying gram, splitting of gram indicate optimal Drying.
2. Bullocks pulling a heavy stone roller are allowed to trample over the harvested black gram crops, spread out in the threshing yard so as to separate the grains.
3. Yield will be higher in black gram crop, if it is sown in the second fortnight of September
4. Neem oil is sprayed @ 6 lit/acre, to control powdery mildew in black gram crop.
5. Mixing of blackgram seeds with ash and storing them in earthen pots for longer period (Also for cowpea and greengram)
6. Coating the black gram with castor oil to increase the keeping quality
7. Mixing the black gram with sweet flag (*Acorus calamus*) powder for seed purpose.
8. Blackgram grains broken into halves will escape from weevil attack during storage.

COWPEA

1. Putrefied buttermilk is sprayed on cowpea crop to control yellow mosaic disease.(Also for Greengram)
2. Vegetable oil is mixed with cowpea before storage.
3. For safe storage, cowpea seeds are filled in earthen pot to its 4/5th volume and the remaining volume is filled with ash. (Also for field bean)
4. Mix cowpea seeds with red earth slurry, dry and store them in earthen pots for one year

GREEN GRAM

Mixing the green gram seeds with sand before storage

PEAS

1. Raising the peas during January i.e. during heavy mist time reduces the plant growth and leads to poor Yield.
2. Three harvestings with 10 days interval to ensure good harvest.

GROUNDNUT

1. Cowpea grown as an intercrop in groundnut fields acts as a trap crop to control red hairy caterpillar.
2. Groundnut pods are threshed by spreading on a threshing floor and beating with wooden sticks (for oil extraction)
3. Kitchen ash is spread around the groundnut storage bags to prevent insect attack
4. Sandy soil is highly suitable for groundnut crop since it gives less number of ill-filled pods.
5. Grow cowpea as border crop to attract Red Hairy Caterpillar.
6. During nights, burning heaps of straw in several places near the fields along with a bucket full of water or castor cake dissolved water near the fire to attract and kill the insects.
7. Spray lime solution to control leaf roller.
8. Spray water to control leaf roller attack.
9. Grow castor as a border crop (trap crop) to reduce the attack of tobacco cut worms.
10. Summer Ploughing is practiced to expose and destroy the pupae of red hairy caterpillar pest
11. Grinding well and dissolving 10 kg of the leaves of *Aloe vera* in water and for an acre to control Red Hairy Caterpillar.
12. To control groundnut ring mosaic, dried sorghum or coconut leaves are powdered and boiled in water to 60°C for one hour, filtered, diluted and sprayed for two times at 10 and 20 days after sowing.
13. Mix neem oil with irrigation water during second or third irrigations to prevent root rot in groundnut.

14. Spray neem oil @6 lit /acre to control root rot.
15. Neem oil solution 4% or neem kernel extract 6 is sprayed to c

SUNFLOWER

1. Rubbing the nearby two flower heads for better seed set.
2. Application of cow dung before planting ensures bigger size sunflower head.

COTTON

1. For delinting cotton seeds, the seed with wet cow dung and dried for 30 minutes and then slightly rub against a stone and clean.
2. Coating the cotton seeds with red soil and sun drying them before sowing to ensure good germination and to make the dibbling easy
3. Mixing one kg of cotton seeds with 200 ml of neem oil und pasting it with fresh cow dung and then drying this over night before sowing to avoid pests.
4. Applying groundnut cake three times during the crop period to enhance plant growth and boll reduction.
5. Growing castor as intercrop or border crop to attract tobacco cut worms
6. Maize and lady's fingers are also grown as intercrops to reduce boll worm attack.
7. A mixture of fish meal and Jaggery at 2:1 ratio is broadcasted on cotton fields to attract cranes and bulbuls, which feed on tobacco cut worms.
8. 5 kg of neem kernels are powdered and soaked in 100 lit of water for one day and filtered in the next day, diluted, mixed with soap solution and sprayed to control white flies.
9. One meter of white cloth is cut in to 10 pieces, soaked in turmeric solution and dried. They are again soaked in castor oil and tied in the fields in different locations under which chimney lamps are kept to control sucking pests like white flies
10. About 600 g. of tobacco is soaked in water for 2-3 days, filtered and sprayed to control white flies.
11. Sugar solution and neem oil are mixed with water and sprayed to control mealy bugs
12. Dusting ash on cotton leaves to control aphids/thrips
13. Cultivating sorghum or pearl millet around cotton fields to prevent whiteflies and thrips
14. Powdering the neem kernels with 'Vasambu' (Sweet flag- *Acoruscalamus*), soaking in water even night and spraying the filtrate in the next day to control all pests.
15. Kapas (lint + seed) are picked up only during morning hours to avoid dust

SUGARCANE

1. Tank silt is applied before planting for higher yield
2. Sugar content of the cane is increased if sheep penning is practiced or sheep manure is applied

3. Detrashing the canes to control scales and mealy bugs. It is locally called as 'SogaiUriththal
4. Providing frequent irrigation to control termites
5. Chulah ash is applied 2-3 months after planting to control early shoot borer
6. Neem kernels or neem cake are powdered, soaked in water overnight and the filtrate is diluted and sprayed to control early shoot borer
7. Dried neem fruits are powdered and applied a 200kg/ha. to control stem borers and fungal diseases.
8. Jaggery (country sugar) is manufactured indigenously. Sugarcane juice is poured in to a wide mouthed big iron utensil called 'Kopparai" and boiled for 1hour. Then baking soda is put to remove the dusts and continuously boiled for another hour with addition of colouring agent. After reaching requir consistency of sticky, colloidal nature (Pagu), juice is poured in to moulds or moulded by hands.

PLANT PROTECTION THROUGH INDIGENOUS TRADITIONAL KNOWLEDGE

1. Application of buttermilk @ 5 lit./40 Kg seed of pulses viz. lentil, arhar, chickpea to protect the crop from wilt disease
2. Farmers broadcast 8 kg common salt to protect the paddy crop from blight disease
3. Dry neem leaves and kanja leaves are used by the farmers to store the grain like wheat, rice etc
4. Mixed cropping of arhar & jowar to protect the arhar from wilt disease
5. Planting of marigold after 8-10 lines of tomato or chilli to protect the crop from mosaic and nematode
6. The farmers practice putting of twigs in field crops for sitting of owl to control the rats
7. Mixed cropping like arhar+ urd+ til+bajra is practiced.
8. Spreading of cooked rice in field crop attract the birds who eat hairy catter pillar
9. The farmers using to control the aphids spray water in mustard crop
10. Ash dusting is practiced in Zaid vegetables to control the red beetle
11. Cutting of upper portion of leaves before transplanting of paddy to prevent the attack of stem borer
12. Spray of extract of 1 kg garlic + 250 gim tobacco + 250 gmgur in 20 lit. to control Gandhi bugs in rice
13. Farmers practice mixed cropping of onion + sugarcane to control the shoot borer in sugarcane
14. Mixed cropping of coriander + chickpea, protects chickpea from pod borer
15. Intercropping of marigold control nematodes in brinjal/tomato.
16. Gram seed treated with heang (Asafoetida) @ 10 gm/10 Kg seed control pod borer and wilt

17. Also Application of 100 gm mustard Oil /40 Kg of gram to control wilt disease.
18. Soaking of seed in buttermilk, (@ 5 lit/40 kg seed) control wilt disease in Pigeonpea and gram.
19. Farmers spray dung solution around field and border area of crops (as repellent) for saving their crops from damage caused by bull.
20. Solution containing 10 lit of cow urine + 1.5 Kg gum + 1.5 kg mustard oil kept for ten days in shed then added 200 lit of water and spread on paddy crop to prevent various diseases& pest.
21. Farmers are broadcasting ash to get higher yield of berseem.
22. To develop yellow colour in mango, farmers are using, 1gm turmeric powder mixed with 50gm carbide for 10 kg mango during the process of ripening.
23. Farmers are applying 8-12 kg common salt per acre with seed to control the wilt in pulse crops
24. Mixed cropping of coriander with gram and linseed with lentil control pod borer and wilt respectively
25. Vegetable seeds are stored by mixing with sawdust of pine.
- 26 Ash is used at the time of sowing of potato to minimize the soil borne diseases
27. Inter-cropping of groundnut with bajra controls nematodes.
28. Extract of garlic 3 kg + tobacco 250 gm + 500 gmNeem leaf is used by the farmers to control the yellow mosaic in vegetables.

Lecture No. 10

Crop Voyage In India And World

Voyage: A long journey especially by ship.

A long journey involving travel by sea or in space

A passage through air or space, as a flight in an airplane or space vehicle

Crop Voyage : A course of travel or passage of crops, especially a long journey by water to a distant place.

A journey of crops from one place to another by land

Crop have very limited mobility and consequently rely upon a variety of medium to transport their propagates including both abiotic and biotic vectors. Seeds can be dispersed away from parent crop or collectively, as well as dispersed in both space and time. The patterns of seed dispersal are in large part by the dispersal mechanism and this has important implications for the demographic and genetic structure of plant populations, as well as migration patterns and species interactions. There are five main modes of seed dispersal: gravity, wind, ballistic, water and by animals. Some plants are only dispersing their seeds in response to an environmental stimulus.

Agriculture is one of the most momentous inventions of humanity. It generated a surplus of food as a consequence of which a sizable fraction of societies were able to engage in economic activities other than agriculture. Thus, agriculture is considered to have been a prerequisite for the development of many civilizations throughout the world. It originated independently and at similar times (around 10000 years ago) in different areas on several continents, notably South East Asia (India), South America, Near East and China. Selection resulting from the cultivation and harvesting of plants by humans induced genetic modifications in these plants resulting in their domestication. Crop plants are unable to disperse their seeds and lack seed dormancy. They also have smaller plants but larger harvested organs (fruits, seeds, leaves) and contain lower amounts of toxic compounds. Flowering time is also less susceptible to environmental influences, mainly daylength. The crop voyage in any agricultural region is actually an assemblage originating throughout the world. During their evolution under cultivation, plants have been subjected to their reduction in genetic diversity. Therefore, wild progenitors and primitive cultivars in the areas of origin are a source of additional genetic diversity to further improve elite cultivars of our major crop plants.

The knowledge of the crop voyage is mainly important in order to avoid genetic erosion, the

loss of germplasm due to the loss of ecotypes and landraces, loss of habitat (such as rainforests), and increased urbanization. Germplasm preservation is accomplished through gene banks (largely seed collections but now frozen stem sections) and preservation of natural habitats (especially in centers of origin).

World centers of origin of cultivated plants (Crop voyage in India and world)

Center	Plants
1) South Mexican & Central American centre	Includes southern sections of Mexico, Guatemala, Honduras and Costa Rica Grains and Legumes: maize, common bean, lima bean, tepary bean, jack bean, grain amaranth Melon Plants: Malabar gourd, winter pumpkin, chayote Fiber Plants: upland cotton, bourbon cotton, henequen (sisal) Miscellaneous: sweet potato, arrowroot pepper, papaya, guava, cashew, wild black cherry, cochineal, cherry tomato, cacao
	62 plants listed three sub centers
2) South American Center	2) Peruvian, Ecuadorean, Bolivian Center Root Tubers: Andean potato, Other endemic cultivated potato species. Fourteen or more species with chromosome -numbers varying from 24 to 60, Edible nasturtium Grains and Legumes; starchy maize, lima bean, common bean Root Tubers: edible canna, potato Vegetable Crops: pepino, tomato, ground cherry, pumpkin, pepper Fiber Plants: Egyptian cotton Fruit und Miscellaneous: cocoa, passion flower guava, quinine tree, tobacco, cherimoya, coco 2 A) Chiloe Center (Island near the coast of southern Chile) Common potato (48 chromosomes) Chilean strawberry 2B) Brazilian-Paraguayan Center manioc, peanut, rubber tree, pineapple, Brazil nut, cashew, Erva-mate, purple grandilla
3) Mediterranean Center	Includes the borders of the Mediterranean Sea. 84 listed plants Cereals and Legumes: durum wheat, emmer, Polish wheat, spelt, Mediterranean oats, sand oats, canary grass, grass pea, pea, lupine Forage Plants: Egyptian clover, white clover, crimson clover, serradella Oil and Fiber Plants: flax, rape, black mustard, olive Vegetables: Garden beet, cabbage, turnip, lettuce, celery, chicory, parsnip, rhubarb Ethereal Oil and Spice Plants: caraway, anise, thyme peppermint, sage, hop.
4) Middle East	Includes interior of Asia Minor, all of Transcaucasia, Iran, and the highlands of Turkmenistan. 83 species Grains and Legumes: eninkorn wheat, dunum wheat, poulard wheat, common wheat, oriental wheat, Persian wheat, two-row barley, rye Mediterranean oats, common oats, lentil, lupine

	Forage Plants: alfalfa. Persian clover, fenugreek, vetch, hairy vetch Fruits: fig, pomegranate. apple quince, cherry, hawthorn
5) Ethiopia	Includes Abyssinia, Eritrea, and part of Somaliland. 38 species listed: rich in wheat and barley. Grains and Legumes: Abyssinian hard wheat, poulard wheat, emmer, Polish wheat, barley, grain sorghum, pearl millet, African millet, cowpea, flax, teff Miscellaneous: Sesame, castor bean, garden cress, coffee, okra, myrrh indigo
6) Central Asiatic Center	Includes Northwest India (Punjab, North west Frontier Provinces and Kashmir) Afghanistan, Tajikistan Uzbekistan and western Tian-Shan. 43 plants Grains and Legumes: common wheat club wheat, shot wheat, peas, lentil, horse bean, chickpea, mug, mustard, flax, sesame Fiber Plants: hemp, cotton Vegetables: onion. garlic, spinach carrot Fruits: pistachio, pear, almond, grape, apple,
7) Indian Center	Two sub centers Indo-Burmai Main Center(India): includes Assam and Burma, but not Northwest India, Punjab nor Northwest Frontier Provinces 117 plants Cereals and Legumes: rice, chickpea pigeon pea, urd bean, mung bean, rice bean, cowpea Vegetables and Tubers: eggplant, cucumber, radish taro, yam Fruits: Mango, orange, tangerine, citron, tamarind Sugar, Oil and Fibre Plants: Sugarcane, coconut palm, sesame safflower, tree, cotton, oriental cotton, jute, crotalaria, kenaf Spices, Stimulants, Dyes and Miscellaneous: hemp, black pepper, gum arabic, Sandal wood, indigo, cinnamon tree, croton, bamboo 7 A) Siam-Malaya-Java: statt Indo-Malayan Center: Includes Indo-China and the Malay Archipelago, 55 plants Cereals and legumes: Job's tears, velvet bean Fruits; pummel, banana, breadfruit, mangos teen Oil, Sugar, Spice, and Fiber Plants: candlenut, coconut palm sugarcane, clove nutmeg, black pepper, manila hemp
8) Chinese Center	A total of 136 endemic plants are listed in the largest independent center Cereals and Legumes: e.g. broomcorn millet, Italian millet, Japanese barnyard millet, Kaoliang buckwheat, hull-less barley, soybean, Adzuki bean. velvet bean Roots, Tubers, and Vegetables: e.g Chinese yam, radish, Chinese cabbage, onion, cucumber Fruits and Nuts: e.g. pear, Chinese apple, peach, apricot cherry, walnut, litchi Sugar, Drugs and fibre plants: e.g. Sugar, opium poppy, ginseng, camphor, hemp

Rice

- ❖ Most important tropical cereal and supplies a quarter of the entire caloric intake of the human race.
- ❖ About 90% of its area and consumption is in South and Southeast Asia,
- ❖ It belongs to the genus *Oryza* and there are two main cultigens, i.e., *sativa* in Asia and *glaberrima* in Africa.
- ❖ Aaquatic rice species, i.e., *Zizania aquatica* and *Z. palustris*, are endemic to North America, where it is the staple food of Indians.

Origin

- ❖ The Himalayas and adjoining Asia mainland (from Assam, Bangladesh, Burma, Thailand, southern China, and northeren Vietnam).
- ❖ Asian rice culture was established around 7000 years ago.
- ❖ In India carbonized grains excavated from Hastinapur (New Delhi) during 1100-800 BC.
- ❖ During excavation of rice samples belonging to 5000-4000 BC in Thailand that from this place rice spread to other countries

Evolutionary history

- ❖ Genus *Oryza* arose from a common ancestor.
- ❖ In *oryza sativa*, the evolution of different geographical races, i.e., *japonica*, *javanica*, and *indica* by human selecton.
- ❖ In rice, change might have occurred in the following sequence: Perennial->climatic stress->seasonal->human selection-> cultivated rice.

Cotton

Agri-history of Cotton in India :

- ❖ Indian subcontinent has been traced to the 4th millenium BC.
- ❖ The fabrics dated approximately 3000 BC, recovered from the Mohenjo-daro excavations in Sind (Pakistan), were identified to have originated from cotton plants, closely related to the *Gossypium arboreum* species.
- ❖ The lint-bearing species of the genus *Gossypium*, the true cottons, are four, out of which the diploid ($2n=26$) species *G. arboreum* and *G. herbaceum* are indigenous in Asia and Africa.
- ❖ 18th century AD : Introduction of the new world cottons (tetraploid species of *G. hirsutum* and *G. barbadense* with $2n=52$) into India .
- ❖ Last decade of the 20th century (1996-97): India- pride of place in the global cotton statistics (largest cropped area of 8.9 million).

Origin

- ❖ Indigenous cottons
- ❖ The cotton textiles of the Harappan civilization (2300-1750BC) were produced by sophisticated textile craftsmanship.
- ❖ Wild and weedy types associated with old world species of *G. herbaceum* and *G. arboreum*.
- ❖ *G. herbaceum*: northwest of Karachi (Pakistan), northern Baluchistan to south Yemen, Ethiopia, and Sudan and even in West Africa.
- ❖ *G. arboreum*: in Kathiawar, Gujarat, Khandesh, and the Deccan in India. It was in Gujarat (India) or Sind (Pakistan)
- ❖ Three perennial races of *G. arboreum*: *G. burmanicum* of northeastern India, *G. indicum* of western India and the Peninsula, and *G. sudanense* of northern Africa,
- ❖ each contributed separately to the cultivated cottons in Asia and Africa

History of cotton development

Middle of the 18th century: only indigenous arboreum and herbaceum varieties of cotton were grown in different regions of the country.

In 1788, the Governor General (at Calcutta) was requested by London to encourage growth and improvement of Indian cottons to meet the requirements of the Lancashire textile industry. The figures for exact area under indigenous cottons and production in India during this period are not available, although it is reported that the local production had stabilized by 1900 AD.

Sugarcane

- ❖ The origin of sugarcane was India.
- ❖ *Saccharum officinarum* was first domesticated in India and the spread to other countries by Arab merchants.
- ❖ 3000-7000 years ago, (Atarna veda): It is originated from the area Sakkaram and then later it was indicated as sakkra in Sanskrit.

Major plant species domesticated, introduced and cultivated in India.

Crops domesticated in Indian sub-continent

Cereals	<i>Oryza sativa</i> (Rice)
Millets and Forages	<i>Cenhrus ciliaris</i> (bunch grass), <i>Coix lacrymz-jobi</i> (job's tears), <i>Digitaria cruciata</i> (Digitaria), <i>Echinochloa colona</i> (Deccan grass), <i>Echinochloa crus-galli</i> (cockspur grass), <i>Panicum arntidotale</i> (blue panicum), <i>Panicum miliaceum</i> (french millet), <i>Panicum sumatrense</i> (little millet), <i>Paspalum scrobiculatum</i> (kodo-millet), <i>Sesbania bispinosa</i> syn. <i>Sesbania aculeata</i> (dhaincha)

Grain legumes	<i>Cajanus cajan</i> (pigeonpea), <i>Dolichos biflorus</i> syn. <i>Macrotyloma uniflorum</i> (horse gram/kulthi), <i>Dolichos lablab</i> (hyacinth bean), <i>Mucuna utilis</i> (velvet bean), <i>Psophocarpus tetragonolobus</i> (Goa bean), <i>Vigna aconitifolia</i> (moth bean), <i>Vigna angularis</i> (adzuki bean), <i>Vigna mungo</i> (black gram), <i>Vigna radiata</i> (green gram), <i>Vigna triloba</i> (jungi bean), <i>Vigna umbellata</i> (rice bean)
Oilseeds	<i>Brassica juncea</i> (indian mustard), <i>Brassica rapa</i> ssp. <i>trilocularis</i> (yellow sarson or Indian Colza), <i>Brassica rapa</i> var. <i>toria</i> (Indian rape), <i>Brassica rapa</i> var. <i>dichotoma</i> (brown sarson), <i>Sesamum indicum</i> (sesame)
Fibre crops	<i>Agave cantala</i> (kantala), <i>Bombax malabaricum</i> (red silk cotton), <i>Corchorus capsularis</i> (white jute), <i>Corchorus olitorius</i> (tossa jute), <i>Crotalaria juncea</i> (sun-hemp), <i>Gossypium arboreum</i> (tree-cotton), <i>Gossypium obusifolium</i> (Asiatic cotton), <i>Hibiscus cannabinus</i> (Deccan hemp), <i>Hibiscus sabdariffa</i> (Jamaica sorrel)
Vegetables	<i>Abelmoschus esculentus</i> (okra), <i>Amaranthus blitum</i> (sag chulai), <i>Amaranthus tricolour</i> (lal sag), <i>Amorphophallus campanulatus</i> (elephant-foot yam), <i>Chenopodium album</i> (common pigwheat), <i>Coccinia indica</i> (koi fruit), <i>Colocasia esculenta</i> (taro), <i>Cucumis sativus</i> (cucumber), (Indian lett) (ridge gourd), <i>Luffa cylindrica</i> (sponge gourd), <i>Momordica charantia</i> (bitter gourd), <i>Momordica cochinchinesis</i> (kakora), <i>Momordica dioica</i> (balsam apple), <i>Moringa oleifera</i> (drumstick), <i>Praecitrullus fistulosus</i> (round gourd), <i>Raphanus caudatus</i> (rat-tail Radish), <i>Rumex vesicarius</i> (bladder dock), <i>Sesbania grandiflora</i> (agathi), <i>Solanum melongena</i> (brinjal), <i>Spinacia oleracea</i> (spinach), <i>Trichosanthes anguina</i> (snake gourd), <i>Trichosanthes dioica</i> (pointed gourd)
Fruits	<i>Aegle marmelos</i> (Bengal quince), <i>Areca catechu</i> (areca nut), <i>Artocarpus heterophyllus</i> (jack fruit), <i>Carissa congesta</i> (karaunda), <i>Citrus limon</i> (lemon), <i>Mangifera indica</i> (mango), <i>Musa paradisiaca</i> (plantain/banana), <i>Phoenix sylvestris</i> (date sugar palm, Indian palm), <i>Zizyphus jujuba</i> (Indian jujube), <i>Zizyphus nummularia</i> (wild jujube)
Some important medicinal and	<i>Abelmoschus moschatus</i> (muskmallow), <i>Atropa acuminata</i> (Indian belladonna), <i>Azadirachta indica</i> (margosa tree), <i>Cassia fistula</i> (Indian laburnum), <i>Cymbopogon marini</i> (palmarosa), <i>Cymbopogon nardus</i> (citronella grass), <i>Cymbopogon pendulus</i> (lemon grass), <i>Cynodon dactylon</i>

aromatic plants	(Bermuda grass), Datura metel (datura), Emblica officinale (Indian gooseberry), Patchouli (patchouli), Rauvolfia serpentina (serpentine root), (costus), Vetiveria zizanioides (vetiver)
Narcotics Spices and condiments	Cannabis sativa (hemp) Cinnamomum tamale (Indian cassia) Cinnamomum verum (cinnamon), Crocus sativus (saffron), Curcuma amada (mango ginger), Curcuma caesia (black turmeric), Curcuma domestica syn. C. longa (turmeric), Curcuma zedoaria (zedoary), Elettaria cardamomum (small cardamom), Kaempferia galanga (chandramula), Mentha piperita (mint), Murraya koenigii
Spices and Condiments	(curry leaf tree), Myristica malabaricum (nutmeg), Piper betel (betel pepper), Piper longum (long pepper), piper nigrum (black pepper), Trigonella foenum-graecum (fenugreek), Zingiber officinale (ginger)
Others	Acacia catechu (kattha), Acacia nilotica (India acacia), Bambusa arundinacea (thorny/spiny bamboo), Bambusa tuldo (Bengal bamboo), Bauhinia purpurea (camel foot tree), Borassus flabellifer (palmyra-palm), Caesalpinia sappan (Indian redwood), Camellia sinensis var. assamica (tea), Cedrela toona (red cedar), Cordia myxa (Indian cherry), Coffea bengalense (coffee), Curcum angustifolia (Indian arrowroot), Dendrocalamus hamiltonii (Dendrocalamus), Dendrocalamus strictus (Calcutta bamboo), Dioscorea alata (greater yam), Dioscorea esculenta (lesser yam), Ficus bengalensis (banyan tree), Ficus elastica (Indian rubber), Ficus religiosa (peepal), Garcinia sylvestris (wild mangosteen), Indigofera tinctoria (indigo), kochia indica (bui), Lawsonia alba (henna), Maoutia puya (pua, poi), Marsdenia tinctoria (rion), Morinda angustifolia (ban haldi), Morinda citrifolia (Indian mulberry), Nephelium longana (anshaphal), Nerium indicum (kaner), Nyctanthes arbortristis (Tree of Sadness), Ochlandra travancorica (elephant grass), Oroxylum indicum (sonapatha), Pluchea indica (pludina), Rubia cordifolia (Indian madder), Saccharum officinarum (sugarcane), Saccharum sinense (sugarcane), Sapindustrifoliatus (soap nut tree), Sida rhombifolia (Cuba jute), Sinocalamus giganteus (Sinocalamus), Tamarindus indica (tamarind)

Crops introduced by Portuguese

Pseudo cereals	Amaranthus caudatus (amaranth)
Oilseeds	Arachis hypogaeae (groundnut)
Vegetables	Cucurbita moschata (pumpkin), Ipomoea batatas (sweet potato), Solanum tuberosum (potato)
Fruits	Anacardium occidentale (cashew nut), Anona squamosa (custard apple), Psidium guajava (guava)
Narcotics	Nicotiana tabaccum (tobacco)
Spices and condiments	Capsicum annuum (chilli)

Crops introduced by Britishers

Pseudo cereals	Avena sativa (oat)
Grain legumes	Castanospermum australe (black bean), Pisum sativum (pea)
Fiber crops	Gossypium barbadense (cotton)
Vegetables	Allium tuberosum (leek), Asparagus racemosus (satawar), Beta vulgaris (beet root), Brassica oleracea var. botrytis (cauliflower), Brassica oleracea var. gemmifer (Brussels pekinensis (celery), Capsicum frutescens (sweet pepper), Cichorium intybus (chicory), Cucurbita maxima (squash), Daucas carota (carrot, orange type), Lactuca sativa (lettuce), Lycopersicon esculentum (tomato), Pisum sativum (sweet pea)
Fruits	Averrhoa bilimbi (bilimbi), Averrhoa carambola (carambola), Carica papaya (papaya), Eugenia jambos (rose apple), Fragaria ananassa (strawberry), Garcinia mangostana mangosteen), Helianthus tuberosus (artichoke), Manihot esculenta (cassava), Malus pumila (apple), Prunus armeniaca (apricot), Prunus avium (cherry), Prunus communis syn. P. domestica (plum), Prunus persica (peach), pyrus communis (pear), Ribes rubrum (red currant)
Medicinal and aromatic plants	Cinchona officinalis (quinine), Origanum vulgare (majoram), Papaver somniferum (opium poppy), Pelargonium capaitatium (Geranium), Salvia officinalis (sage), Thymus vulgaris (thyme), Vanilla aromatica (vanilla)

Crops introduced from West and Central Asia by Mughals or Arabs

Allium cepa (onion), Allium sativum (garlic), Brassica rapa (turmip), Brassica oleracea var. capitata (cabbage), Coriandrum sativum (coriander), Cucumis melo (sweet muskmelon), Daucas carota (carrot, black & red type), Phoenix dactylifera (date palm), Pisum sativum (pea), Syzygium aromaticum (clove), Vitis vinifera (grape)

Crops introduced from China

Aleurites fordii (tung-oil), Glycine max (soyabean), Eriobotrya japonica (loquat), Juglans regia (walnut), Litchi chinensis (litchi), Sapium sebiferum (tallow-tree)

Crops introduced from Latin America: *Hevea brasiliensis* (Rubber), *Ananas comosus* (pineapple)

Crops introduced from Southeast Asia and Pacific islands

Arenga pinnata (sugar-palm), *Artocarpus communis* (breadfruit), *Citrus decumanus* (pomelo), *Citrus paradisi* (grapefruit), *Durio zibethinus* (durian) and *Metroxylon sagu* (sago)

Some recent introductions

Humulus lupulus (hops), *Helianthus annuus* (sunflower), *Simarouba glauca* (simarouba), *Cyphomandra betacea* (tree tomato), *Carya illinoensis* (pecan nut), *Corylus avellana* (hazel nut), *Macadamia tetraphylla* (macadamia nut), *Parthenium argentatum* (guayule), and *Mentha arvensis* (spearmint, USA) *Acacia senegal* (Australia), *Acacia mangium* (Australia) and *Actinidia chinensis* (kiwifruit), (New Zealand).

Lecture No 11

Agriculture scope in India and Maharashtra, Chronological agricultural technology development in Indian Agriculture

Define the agriculture and state the how it is art, science and business of crop production.

Agriculture

The term Agriculture is derived from two *Latin* words **ager** or **agri** meaning **soil** and **cultura** meaning **cultivation**.

Agriculture is an applied science which encompasses all aspects of crop production including horticulture, livestock rearing, fisheries, forestry, etc.

Agriculture is defined as an **art, science and business** of producing crops and livestock for economic purposes.

As an art

- It embraces knowledge of the way to perform the operations of the farm in a skillful manner, but does not necessarily include an understanding of the principles underlying the farm practices.

As a science:

- Utilizes all technologies developed on scientific principles such as crop breeding, production techniques, crop protection, economics etc. to maximize the yield and profit.
- For example, new crops and varieties developed by
 1. Hybridization,
 2. Transgenic crop varieties resistant to pests and diseases,
 3. Hybrids in each crop,
 4. High fertilizer responsive varieties,
 5. Water management,
 6. Herbicides to control weeds,
 7. Use of bio-control agents to combat pest and diseases etc.

As the business:

- As long as agriculture is the way of life of the rural population production is ultimately bound to consumption.
- But agriculture as a business aims at maximum net return through the management of land labour, water and capital, employing the knowledge of various sciences for production of food, feed, fibre and fuel.
- In recent years, agriculture is commercialized to run as a business through mechanization.

Scope and Importance of Agriculture in India and Maharashtra

Scope of Agriculture

The scope of the agriculture is very vast in India and Maharashtra state as it is most important enterprise. In Indian economy agriculture provides food for population fodder to livestock and fulfils other needs like fiber , fuel, timber ,wood and raw materials to various industries. About 60% Indian population is engaged in agriculture and agro based industries . At present agriculture contributes about 35% share of the Indian economy. India is a exporter of fruits, tea, coffee, basmati rice and other agriculture commodities which help in getting foreign exchange. Therefore, agriculture is considered as backbone of the Indian economy. In the world India is only country having all types of soil and climatic conditions suitable for growing variety of crops . If proper attention is being paid to exploit, all the agricultural resources and technology developed in the field of agriculture, India will be at the top in the world.

In Maharashtra four agriculture universities and state department of agriculture are engaged with the job of education, research and transfer of technology in the field of agriculture to address various problems like low soil fertility crop varieties of low genetic yield , control of diseases and insects, non availability of production inputs government economic policies affecting agriculture and research and extension programs. It is necessary to increase agriculture production from unit area and to adopt intensive agriculture in order to increase national production.

Proverbially, India is known as “Land of Villages”. Near about 67% of India’s population live in villages. The occupation of villagers is agriculture. Agriculture is the dominant sector of our economy & contributes in various ways such as:

National Economy:

- Indian Agriculture is one of the most significant contributors to the Indian economy.
- Agriculture is the only means of living for almost 60% of the employed class in India.
- The agriculture sector of India has occupied almost 43% of India's geographical area.
- Agriculture is still the only largest contributor to India's GDP (16%) even after a decline in the same in the agriculture share of India. Agriculture also plays a significant role in the growth of socio-economic sector in India.
- The per capita income of the agriculture sector declines to 1/3 of the national per capita income during the recent years.
- The per capita income of the agriculture population is estimated around Rs. 10,865 in 2010, which is around 32% of the national per capita income at Rs. 33,802/-.
- Per capita income of the agriculture population was around half (1/2) at Rs. 5,505 of the national per capita income at Rs. 11,433/- during 1980 however, it came down to around 42% at Rs. 6,652/- of the national per capita income at Rs. 16,020/- during 2000.

Total Employment:

- Around 65% population is working & depends on agriculture and allied activities.

- Nearly 70% of the rural population earns its livelihood from agriculture and other occupation allied to agriculture.
- In cities also, a considerable part of labour force is engaged in jobs depending on processing & marketing of agricultural products.

Industrial Inputs:

- Most of the industries depend on the raw material produced by agriculture, so agriculture is the principal source of raw material to the industries.
- The industries like cotton textile, jute, paper, sugar depends totally on agriculture for the supply of raw material.
- The small scale and cottage industries like handloom and power loom, ginning and pressing, oil crushing, rice husking, sericulture fruit processing, etc are also mainly agro based industries.

Food Supply:

- During this year targeted food production was 198 million tons & which is to be increased 225 million tons by the end of this century to feed the growing population of India i.e. 35 corer in 1951 and 100 corers at the end of this century.
- India, thus, is able to meet almost all the need of its population with regards to food by develop intensive program for increasing food production.

State Revenue:

- The agriculture is contributing the revenue by agriculture taxation includes direct tax and indirect tax.
- Direct tax includes land revenue, cesses and surcharge on land revenue, cesses on crops & agril income tax.
- Indirect tax induces sales tax, custom duty and local octri, etc. which farmer pay on purchase of agriculture inputs.

Trade:

- Agriculture plays and important role in foreign trade attracting valuable foreign exchange, necessary for our economic development.
- The product from agriculture based industries such as jute, cloth, tinned food, etc. contributed to 20% of our export.
- Around 50 % of total exports are contributed by agril sector. Indian agriculture plays and important role in roads, rails & waterways outside the countries.
- Indian in roads, rails and waterways used to transport considerable amount of agril produce and agro based industrial products.

- Agril products like tea, coffee, sugar, oil seeds, tobacco; spices, etc. also constitute the main items of export from India.

Source of Livelihood

The main source livelihood of many people is agriculture. Approximately 70 % of the people directly rely on agriculture as a mean of living. This high percentage in agriculture is as a result of none development of non-agricultural activities to absorb the fast-growing population. However, most people in developed countries do not engage in agriculture.

Marketable Surplus

The growth of agricultural sector contributes to marketable surplus. Many people engage in manufacturing, mining as well as other non- agricultural sector as the nation develops. All these individuals rely on food production that they might meet from the nation's marketable surplus. As agricultural sector development takes place, production increases and this leads to expansion of marketable surplus. This may be exported to other nations.

Food Security

A stable agricultural sector ensures a nation of food security. The main requirement of any country is food security. Food security prevents malnourishment that has traditionally been believed to be one of the major problems faced by the developing countries. Most countries rely on agricultural products as well as associated industries for their main source of income.

Chronological Agriculture Technology Development in India

- Agriculture started in different parts of world during different period. As India was pastoral country before agriculture was started the development of crop and animal took place, leading to various type of farming system that are now seen in different parts of the country.
- India is the most important contribution to world agriculture is rice, sugarcane, number of legume and tropical fruit like mango and also native of India.
- Indian agriculture predominantly subsistence type. Commercial type like potato, sweet potato, tomato, chillies, groundnut, cashew nut, tobacco.
- The priorities in agriculture changed in different period of time in human history to meet the necessary of life.
- Human development witnessed stone age, bronze age to reach the Modern stage of development.

1. Stone age

2. Chalcolithic culture

3. Early Aryan Culture

4. Vedic Period

5. Buddhist period

6. Magadhan period

7. Medieval period

8. Mughal period

9. British period

1. Stone Age: this age has been further subdivided

1. Paleolithic age

2. Mesolithic age

3. Neolithic age

1. Paleolithic age

- This period ranged between 500,000 and 100,000 years ago.
- Inhabitants of paleolithic age were identical to homo erectus.
- The man in this age was engaged in acquisition of intelligence and skill to prepare various types tools such as choppers, crude scraper, hand axes.
- This culture mainly locates in Jammu and Kashmir, Central India, Maharashtra and Tamil Nadu.

2. Mesolithic age

- The period lies between 10,000 to 7500 B.C.
- The use of tools includes a large variety of blades, axes, scrapers flakes and chopping tools.
- Mesolithic age man was primarily a hunter and food gatherer mainly inhabited in Gujarat, Madhya Pradesh, Uttar Pradesh and West Bengal.

3. Neolithic age

- The period extends from 7800 to 170 B.C.
- In fact this is the period of agriculture revolution.
- The man of this period acquired the art and skill of grinding and polishing of stone implements used in agriculture.
- The inhabitants during the period domesticated both plants and animals.
- The domesticated plants were wheat and barley while the animal was goat, sheep and cattle.

Main feature of Neolithic culture in India

- Economic and technological development in India.
- Use of polished stone axes to clear bushes.
- Hand made pottery for storing food grains.
- Invented textile, weaving and basketry.
- Discovery of milk.
- Cultivation of rice, banana.
- Cultivation of millets and pulses in south India.

Lecture No 12 & 13

Importance of agriculture and agricultural resources available in India i.e. land, weather, irrigation, labour, capital, market etc.

The Importance of Agriculture

For decades, agriculture has been associated with the production of essential food crops. At present, agriculture above and beyond farming includes forestry, dairy, fruit cultivation, poultry, bee keeping, mushroom, arbitrary, etc. Today, processing, marketing, and distribution of crops and livestock products etc. are all acknowledged as part of current agriculture. Thus, agriculture could be referred to as the production, processing, promotion and distribution agricultural products. Agriculture plays a critical role in the entire life of a given economy. Agriculture is the backbone of the economic system of a given country. In addition to providing food and raw material, agriculture also provides employment opportunities to very large percentage of the population. Below are the importance of agriculture:

Source of Livelihood

The main source livelihood of many people is agriculture. Approximately 70 % of the people directly rely on agriculture as a mean of living. This high percentage in agriculture is as a result of none development of non-agricultural activities to absorb the fast-growing population. However, most people in developed countries do not engage in agriculture.

Contribution to National revenue

Agriculture is the main source of national income for most developing countries. However, for the developed countries, agriculture contributes a smaller per cent age to their national income.

Supply of Food as well as Fodder

Agricultural sector provides fodder for domestic animals. Cow provides people with milk which is a form of protective food. Moreover, livestock also meets people's food requirements.

Significance to the International Trade

Agricultural products like sugar, tea, rice, spices, tobacco, coffee etc. constitute the major items of exports of countries that rely on agriculture. If there is smooth development practice of agriculture, imports are reduced while export increases considerably. This helps to reduce countries unfavourable balance of payments as well as saving foreign exchange. This amount may be well used to import other essential inputs, machinery, raw-material, and other infrastructure that is helpful for the support of country's economic development.

Marketable Surplus

The growth of agricultural sector contributes to marketable surplus. Many people engage in manufacturing, mining as well as other non- agricultural sector as the nation develops. All these individuals rely on food production that they might meet from the nation's marketable surplus. As agricultural sector development takes place, production increases and this leads to expansion of marketable surplus. This may be exported to other nations.

Source of Raw Material

The main source of raw materials to major industries such as cotton and jute fabric, sugar, tobacco, edible as well as non-edible oils is agriculture. Moreover, many other industries such as processing of fruits as well as vegetables and rice husking get their raw material mainly from agriculture.

Significance in Transport

Bulks of agricultural products are transported by railways and roadways from farm to factories. Mostly, internal trade is in agricultural products. Moreover, the revenue of the government, to a larger extent, relies on the success of agricultural sector.

Foreign Exchange Resources

The nation's export trade depends largely on agricultural sector. For example, agricultural commodities such as jute, tobacco, spices, oilseeds, raw cotton, tea as well as coffee accounts for approximately 18 % of the entire value of exports of a country. This demonstrates that agriculture products also continue to be important source of earning a country foreign exchange.

Great Employment Opportunities

Construction of irrigation schemes, drainage system as well as other such activities in the agricultural sector is important as it provides larger employment opportunities. Agriculture sector provides more employment opportunities to the labor force that reduce the high rate of unemployment in developing countries caused by the fast growing population.

Economic Development

Since agriculture employs many people it contributes to economic development. As a result, the national income level as well as people's standard of living is improved. The fast rate of development in agriculture sector offers progressive outlook as well as increased motivation for development.

Food Security

A stable agricultural sector ensures a nation of food security. The main requirement of any country is food security. Food security prevents malnourishment that has traditionally been believed to be one of the major problems faced by the developing countries. Most countries rely on agricultural products as well as associated industries for their main source of income.

Agriculture resources available in India

1. Land
2. Weather
3. Irrigation
4. Labour
5. Capital
6. Market

1. Land

- Land means the material and forces which nature gives freely for man aid, in Land and water, in air and light and heat.
- Land use in crop production is known as soil.
- Land or soil is important natural resources in crop production.
- It is immovable, immobile, indestructible and fixed in quantity.
- Land in the form of soil offers medium, space, nutrition, moisture to the crops

2. Weather

Weather is the atmospheric condition of a place for a short duration with respect to its one or more elements.

The average weather conditions, prevalent from one season to another in the course of a year, over a large area are known as **climate**.

Difference between weather and climate

Weather	Climate
Instantaneous physical state of atmosphere at particular place	Normal physical state or generated condition of atmosphere or long term average condition of place.
Weather changes refer to specific instant of time (day or week)	It is generalized over longer span of time and for longer area.
It is expressed in term of numerical values of meteorological element.	It is expressed in term of time averages and area averages of meteorological elements.
Weather is measured in observatory.	This is derived information on regional basis.
Weather is the study of atmospheric conditions for short duration of a limited area.	Climate is the study of the average weather conditions observed over a long period of time for a larger area.
Weather is influenced by anyone of its predominant elements i.e., temperature or humidity	Climate is the collective effect of all its elements.
The weather changes very often	It is more or less permanent
It is experienced over small areas of a country	It is experienced over large area of the continent

A place can experience different types of weather conditions in a year.	A place can experience only one type of climate
Weather is the study of atmospheric conditions for short duration of a limited area.	Climate is the study of the average weather conditions observed over a long period of time for a larger area.

The Elements of weather and climate are:

1. Temperature
2. Pressure
3. Wind
4. Humidity and
5. Precipitation

3. Irrigation

Water resources: Water is an important natural resource. It is vital for increasing the agricultural production. India is rich in water resource and endowed with a net work of great river and vast alluvial basins to hold ground water.

The total annual water is 185 million ha m of which 135 million ha m is through surface of water resources and 50 million ha m through ground water resources and after meeting the need of water for other purposes about 77million ha m would be available for irrigation by 2025 A.D. On the basis of available water resources, we can irrigate an area of 110 million ha by efficient use of irrigation water,. The gross and net irrigated areas is about 65 and 44 million ha only. By creating more irrigation potential without making provision of adequate drainage facility leads to land degradation.

Need of irrigation in India

1. Uncertainty of Monsoon rainfall both in time and place.
2. Irregularity in distribution of rainfall throughout the year.
3. Excessive rainfall causing flood.
4. Draught is an annual event in some areas.
5. India is a land of Rabi Crops. But there is not rainfall in winter months.
6. Some soils need more water.
7. Introduction of H.Y.V seeds and multiple cropping need water throughout the year.

Why Irrigation is Indispensable in India?

India being an agricultural country mostly depends on monsoons rain that is very much uncertain. Therefore, Indian agriculture needs artificial watering of the fields. The artificial way of supplying water to the agricultural farm or field at the right moment in an appropriate volume for the proper growth of the plants in order to get the maximum yields of cultivation is technically called irrigation.

Various factors that have made irrigation indispensable are:

Uncertain monsoon rain:

In India, major portion of the total annual rainfall occurs in the monsoon season and lasts for around four months in a year. But it is uncertain; sometimes it causes flood and also droughts.

Spatial variation in rainfall:

Many parts of our country such as Rajasthan, Punjab, Haryana, north-western part of Gujarat, interior parts of the Deccan plateau receive very low monsoon rain even less than 75 cm. This rainfall deficiency also makes it necessary to provide irrigation to the fields.

Low winter rain:

To grow rabi crops like wheat, pulses, oil-seeds, etc. in long and dry winter irrigation is also necessary.

Lack of moisture retaining capacity of the soils:

Oils like laterite, red soils are poor in moisture retaining capacity and, as such, without irrigation it is impossible to cultivate this type of soils.

Cultivation of high yielding variety crops:

H.Y.V. seeds require regular and adequate water supply unlike any other normal seeds. This can only be achieved by irrigation.

Multiple cropping:

In this method of cultivation more than one crop are cultivated. It requires regular water supply throughout the year and this can be made possible only through irrigation.

4. Labour

It can be physical work or mental work that is done by person with an aim of earning money. It includes work done by farmer, worker, the service of teacher, doctor, actor etc.

Characteristic of labour

- Labour is inseparable for labourer.
- Labour has very weak bargaining power
- Lack of free mobility.
- Supply of labour is independent of demand
- Supply of labour peculiarly changes with the wages

Agriculture classified labour

1. Landless Agriculture labour

- a) Permanent Labour
- b) Temporary labour

2. Small and Marginal farmer

- a) Cultivator labour
- b) Share labour
- c) Lease labour

5. Capital

Capital is not original factor like land but it is the result of man made efforts. Capital is necessary wealth but all wealth is not necessary capital.

Characteristic of capital

- Capital is not free gift of nature, it is result of man made effort.
- Capital is productive, help in enhancing the overall productivity of all the resources employed in the production process.
- Capital is highly mobile
- Capital is supply elastic as its supply can be altered according to need.

6. Market

Market is the place where goods and service are exchanged. Market consists of buyer and sellers with facilities to communicate with each other for transaction of goods and service.

Marketing is the economic process by which goods and service are exchanged between the producer and the consumers and their determined in term of money prices.

Scope of Agriculture Market

The scope of the field marketing can be examined from five angles such as

1. **Producer Interest**
2. **Consumer Interest**
3. **Society Interest**
4. **Traders Interest**
5. **Governmental Role**

Lecture No 14

Crop Classification and Significance

Importance of classifying the Crop Plants

1. To get acquainted with crops.
2. To understand the requirement of soil & water different crops.
3. To know adaptability of crops.
4. To know the growing habit of crops.
5. To understand climatic requirement of different crops.
6. To know the economic produce of the crop plant & its use.
7. To know the growing season of the crop
8. Overall to know the actual condition required to the cultivation of plant.

Bases of Classifying the Crop Plants (CGCLCENcPPBS)

- A. Classification based on climate
- B. Classification Based on growing season
- C. Use/Agronomic classification
- D. Classification based on life of crops/duration of crops
- E. Classification based on cultural method/water
- F. Classification based on root system
- G. Classification based on economic importance
- H. Classification based on No. of cotyledons
- I. Classification based on photosynthesis' (Reduction of CO₂/Dark reaction)
- J. Classification based on length of photoperiod required for floral initiation
- K. Botanical classification
- L. Special purpose classification

A. Classification based on climate

1. **Tropical:** Crops grow well in warm and humid climate e.g Rice, Maize, Sorghum etc.
2. **Temperate:** Crops grow well in cool climate. E.g. Wheat, Oats, Gram, Potato etc.

B. Classification Based on growing season

1. Kharif/Rainy/Monsoon crops

The crops grown in monsoon months from June to Oct-Nov, Require warm, wet weather at major period of crop growth, also required short day length for flowering. E.g. Cotton, Rice, Jowar, bajara.

2. Rabi/winter/cold seasons crops:

Require winter season to grow well from Oct to March month. Crops grow well in cold and dry weather. Require longer day length for flowering. E.g. Wheat, gram, sunflower etc.

3. Summer/Zaid crops:

Crops grown in summer month from March to June. Require warm day weather for major growth period and longer ay length for flowering. E.g. Groundnuts, Watermelon, Pumpkins, Gourds.

C. On the basis of use/Agronomic classification:

1. Grain crops: may be cereals as millets cereals are the cultivated grasses grown for their edible starchy grains. The larger grain used as staple food is cereals. E.g. rice, Jowar, wheat, maize, barley, and millets are the small grained cereals which are of minor importance as food. E.g. Bajara.

2. Pulse/legume crops: seeds of leguminous crops plant used as food. On splitting they produced dal which is rich in protein. E.g. green gram, black gram, soybean, pea, cowpea etc.

3. Oil seeds crops: crop seeds are rich in fatty acids, are used to extract vegetable oil to meet various requirements. E.g. Groundnut, Mustard, Sunflower, Sesamum, linseed etc.

4. Forage Crop: It refers to vegetative matter fresh as preserved utilized as food for animals. Crop cultivated & used for fickler, hay, silage. Ex- sorghum, elephant grass, guinea grass, berseem & other pulse bajara etc.

5. Fiber crops: crown for fiber yield. Fiber may be obtained from seed. E.g. Cotton, steam, jute, Mesta, sun hemp, flax.

6. Roots crops: Roots are the economic produce in root crop. E.g. sweet, potato, sugar beet, carrot, turnip etc.

7. Tuber crop: crop whose edible portion is not a root but a short thickened underground stem. E.g. Potato, elephant, yam.

8. Sugar crops: the two important crops are sugarcane and sugar beet cultivated for production for sugar.

9. Starch crops: grown for the production of starch. e.g. tapioca, potato, sweet potato.

10. Drug crop/Narcotics: used for preparation for medicines. e.g. tobacco, mint, ganja.

11. Spices & condiments/spices crops: crop plants as their products are used to flavor taste and sometime color the fresh preserved food. e.g. ginger, garlic, chili, cumin onion, coriander, cardamom, pepper, turmeric etc.

12. Vegetables crops: may be leafy as fruity vegetables. e.g. Palak, mentha, Brinjal, tomato.

13. Green manure crop: grown and incorporated into soil to increase fertility of soil. e.g. sun hemp, Dhaincha

14. Medicinal & aromatic crops: Medicinal plants includes cinchona, isabgol, opium poppy, senna, belladonna, rauwolfra, iycorice and aromatic plants such as lemon grass, citronella grass, palmorsa, Japanese mint, peppermint, rose geranicem, jasmine, henna etc.

15. Beverages –products are used for mild, agreeable liquors . e.g Tea Coffee

D. Classification based on life of crops/duration of crops:

1. Seasonal crops: A crop completes its life cycle in one season-Karin, Rabi. summer. e.g. rice, Jowar, wheat etc.

2. Two seasonal crops: crops complete its life in two seasons. e.g. Cotton, turmeric, ginger.

3. Annual crops: Crops require one full year to complete its life in cycle. e.g. sugarcane.

4. Biennial crops: which grows in one year and flowers, fructifies & perishes the next year? e.g. Banana, Papaya.

5. Perennial crops: crops live for several years. e.g. Fruit crops, mango, guava etc.

E. Classification based on cultural method/water:

1. Rain fed: crops grow only on rain water. e.g. Jowar, Bajara, Mung etc.

2. Irrigated crops: Crops grows with the help of irrigation water. e.g. Chili, sugarcane, Banana, papaya etc.

F. Classification based on root system:

1. Tap root system: The main root goes deep into the soil. e.g. Tur, Grape, Cotton etc.

2. Adventitious/Fiber rooted: The crops whose roots are fibrous shallow & spreading into the soil. e.g. Cereal crops, wheat, rice etc.

G .Classification based on economic importance:

1. Cash crop: Grown for earning money. e.g. Sugarcane, cotton.

2. Food crops: Grown for raising food grain for the population and & fodder for cattle. e.g. Jowar, wheat, rice etc.

H. Classification based on No. of cotyledons:

1. Monocots or monocotyledons: Having one cotyledon in the seed. e.g. all cereals & Millets.

2. Dicots or dicotyledonous: Crops having two cotyledons in the seed. e.g. all legumes & pulses.

I. Classification based on photosynthesis' (Reduction of CO₂/Dark reaction):

1. C₃ Plants:

Photo respiration is high in these plants. C₃ plants are the most common and the most efficient at photosynthesis in cool, wet climates. C₃ Plants have lower water use efficiency. The initial product of C assimilation in the three 'C' compounds. The enzyme involved in the primary carboxylation is ribulose biphosphate carboxylase/oxygenase causes an oxidation reaction in which some of the energy used in photosynthesis is lost in a process known as photorespiration. The result is about a 25% reduction in the amount of carbon that is fixed by the plant and released back into the atmosphere as carbon dioxide. E.g. Rice, soybeans, wheat, barley cottons, potato.

2. C4 plants:

The primary product of C fixation is four carbon compounds which may be malic acid or aspartic acid. The enzymes responsible for carboxylation are phosphoenolpyruvic acid carboxylase which has high affinity for CO₂ and capable of assimilating CO₂ even at lower concentration, photorespiration is negligible. Photosynthetic rates are higher in C₄ than C₃ plants for the same amount of stomatal opening. These are said to be drought resistant & they are able to grow better even under moisture stress. C₄ plants translocate photosynthates rapidly. C₄ plants are most efficient at photosynthesis in hot, sunny climates. E.g. Sorghum, Maize, Napier grass, sesame-etc.

3. Cam plants:

(Crassulacean acid metabolism plants) the stomata open at night and large amount of CO₂ is fixed as a malic acid which is stored in vacuoles. During day stomata are closed. There is no possibility of CO₂ entry. CO₂ which is stored as malic acid is broken down & released as CO₂. In these plants there is negligible transpiration. C₄ & cam plant have high water use efficiency. These are highly drought resistant. CAM plants are adapted to avoid water loss during photosynthesis so they are best in deserts. e.g. Pineapple, sisal & agave.

J. Classification based on length of photoperiod required for floral initiation:

Most plants are influenced by relative length of the day & night, especially for floral initiation, the effect on plant is known as photoperiodism depending on the length of photoperiod required for floral initiation, plants are classified as

1. Short-day plants: Flower initiation takes place when days are short less than ten hours. E.g. rice, Jowar, green gram, black gram etc.

2. Long day's plants: require long days are more than ten hours for floral initiation. E.g. Wheat, Barley,

3. Day neutral plants: Photoperiod does not have much influence for phase change for these plants. e.g. Cotton, sunflower. The rate of the flowering initiation depends on how short or long is photoperiod. Shorter the days, more rapid initiation of flowering in short days plants. Longer the days more rapid are the initiation of flowering in long days plants.

K. Botanical classification

Division- Spermatophyta

Subdivisions : Angiosperms

(ii) Angiosperms

- Highly specialized plants reproducing by seed within an ovary.
- Include major cultivated crops of economic importance as food, feed and fiber.
- Examples are wheat, cotton, tobacco etc. •

There are two subcategories of angiosperms based on the number of cotyledons in the seed.

(a) Monocotyledons

(b) Dicotyledons

L. Special purpose classification

1. Green manure Crops • Crops grown and ploughed under in the green and mature form in order to improve soil fertility. • Mostly these are legume crops. • Examples are: Dhaincha, berseem.

2. Silage Crops • Crops are cut and preserved as silage in a succulent condition by partial fermentation in silos. • Examples are: Oat, maize soybean, sorghum and grasses.

3. Hay crops • Crops that has been mown and dried for use as fodder. • Examples are: Oat, wheat, sorghum and grasses.

4. Green chop crops • These crops are harvested when still green and succulent and are fed directly to animals without curing. • This is called soiling, zero grazing or green feed. • Examples are: Berseem, lucerne

5. Cover crops • Crops planted to cover the ground and reduce soil erosion and nutrient losses by leaching. • These are usually spreading plants. • Examples are: moth, mash, grasses, rye etc.

6. Catch crops • Crops grown to fill in when the major crops have failed or when planting is delayed for some reason and major crop can not be raised successfully. • Examples are: maize and sorghum for fodder

7. Companion crops • Crops grown together as mixture in same field. • It is also known as polyculture. • Usually legumes are grown mixed with grasses in order to increase forage production and improve forage quality. • Examples are: Berseem and barley, Berseem and oat, maize and lobia etc

8. Relay crops • A relay crop is one which is planted as a second crop after the first crop has reached its reproductive stage of growth but before it is ready for harvest. • Examples are: Sugarcane in sugar beet

Crops Significance

Major crops in India in detail.

Rice



India is the 2nd largest producer of rice in the world. India has largest area in world under rice cultivation.

Type of Crop: Kharif, Rabi

Major Producers(Distribution): West Bengal, Uttar Pradesh, Andhra Pradesh, Punjab, Bihar, Orissa, Chhattisgarh, Assam, Tamil Nadu, Haryana

Highest Producer: West Bengal

Highest productivity: Punjab

Research Centres: Cuttack, Odisha

Highest Producing Country: China

Wheat

It is the 2nd most important food crop in India. It is a Rabi crop. India stands second in production of wheat worldwide.

Type of Crop: Rabi

Major Producers: Uttar Pradesh, Punjab, Madhya Pradesh, Haryana, Rajasthan, Bihar, Gujarat, Maharashtra, West Bengal, Uttarakhand

Highest Producing State: Uttar Pradesh

Highest per Hectare Yield: Punjab

Research Centres: Karnal, Haryana

Highest Producing Country: China



Cotton

Cotton is a tropical and subtropical Kharif crop. It is a fibre crop and is known as 'White gold'. India ranks 3rd in the production of cotton worldwide. It is a dry crop but roots need timely supply of water at maturity.

Type of Crop: Kharif

Major Producers: Gujarat, Maharashtra, Andhra Pradesh, Haryana, Madhya Pradesh, Punjab, Rajasthan, Karnataka, Tamil Nadu, Orissa
Leading Producer: Gujarat (According to the 2015 report)

Research Centres: Nagpur, Maharashtra

Highest Producing Country: China



Jute

Jute is a tropical plant that requires hot and humid climate. It is one of the most important natural fibres in terms of cultivation and usage. Almost 85% of the world's jute is cultivated in the Ganges Delta.

Type of Crop: Zaid

Major Producers: West Bengal, Bihar, Assam, Andhra Pradesh, Orissa, Meghalaya, Nagaland, Tripura, Uttar Pradesh

Largest Producer: West Bengal

Highest per Hectare Yield: West Bengal

Research Centres: Kolkata & Nilgunj, West Bengal

Highest Producing Country: India (but highest exporter is Bangladesh)





Sugarcane Sugarcane is an important cash crop. India stands at 2nd position among all countries in the world in its production. Major Producers: Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Bihar, Gujarat, Haryana, Uttarakhand, Punjab

Largest Producer: Uttar Pradesh

Highest per Hectare Yield: Tamil Nadu

Research Centre: Lucknow

Highest Producing Country: Brazil



Soybean

In India, soybean crop is primarily cultivated in the states of Madhya Pradesh, Rajasthan, Maharashtra, and Uttar Pradesh.



Tea

Tea is an evergreen plant that mainly grows in tropical and subtropical climates. Tea is a labour intensive crop and 50% of the labourers are women. It grows faster under light shade. Commercial cultivation of tea started in India from British era. India is the 2nd largest producer and the largest consumer of tea in the world. Tea plants require high rainfall but its roots

cannot tolerate water logging. Hence, it requires sloppy areas. Major Producers: Assam, Darjeeling (West Bengal), Meghalaya, Kerala, Himachal Pradesh, Tamil Nadu, Karnataka
Highest Producer: Assam

Research Centres: Tocklai, Assam

Highest Producing Country: China

Coffee



Coffees are grown in shade and commonly with two tiers of shade. Growing altitudes of coffee range between 1,000 to 1,500 m above sea level for Arabica (premier coffee), and 500 to 1,000 m for Robusta (lower quality).

Major Producers: Karnataka, Tamil Nadu, Kerala, Andhra Pradesh, Telangana, Odisha, Assam, Meghalaya, Arunachal Pradesh, Manipur, Nagaland

Highest Producer: Karnataka

Highest Producing Country: Brazil

Spices

India has been known for its spices since ancient times.

Cardamom (Queen of Aromatic Spices) – Assam, Kerala, Tamil Nadu

Pepper (King of Spices) – Kerala

Chillies – Andhra Pradesh, Maharashtra, Odisha, Rajasthan

Turmeric – Andhra Pradesh, Karnataka, Tamil Nadu

Nutmeg – Kerala

Areca nut – Kerala, Karnataka, Tripura, Assam

Coconut – Kerala, Tamil Nadu, Karnataka, Andhra Pradesh

Cinnamon – Kerala

Clove – Kerala

Ginger – Kerala, Meghalaya, Sikkim

Farming system is a resource management strategy to achieve economic and sustain agricultural production to meet diverse requirement of the farm household while preserving the resource base and maintaining high environmental quality.

Sustainability is the objective utilization of inputs without impairing the quality of environment with which it interacts. Therefore, it is clear that farming system is a process in which sustainability of production is the objective.

Lecture No 15

National Agriculture Setup In India

Introduction

Agriculture continues to remain a major sector of the Indian economy. It contributes **13.6% of GDP**, provides **58 per cent of employment** and continues to be the primary source of living for 65 per cent of the population.

India's total geographical area is **329 million** hectares. Out of this, 195 million hectare is gross cropped area and 141 million hectare is net sown area.

On the other hand, net irrigated area is only 65.3 million hectares. Rest of the land is rainfed.

National Agricultural Research System (NARS) :

India has built up a fairly advanced agricultural research system.

The most importantly, his work was instrumental for the establishment of the Imperial (now Indian) Agricultural Research Institute in 1905 at Pusa, Bihar. Agricultural Colleges were also established at Pune, Kanpur, Sabour, Nagpur, Coimbatore, and Lyallpur (now in Pakistan).

The ICAR System

- ICAR founded by **16 July 1929** by recommendation of Royal commission of Agriculture.
- The Indian Council of Agricultural Research (ICAR) is an autonomous body responsible for co-ordinating agricultural education and research in India.
- It reports to the Department of Agricultural Research and Education, Ministry of Agriculture.
- The Union Minister of Agriculture serves as its president.
- It is the largest network of agricultural research and education institutes in the world.
- An eminent agricultural scientist was appointed as the Executive Head of the ICAR and was designated as the Director General.
- Director General is assisted by eight Deputy Director General (Subject Matter Divisions) one each in the fields of
- i) Crops, ii) Natural Resources Management iii) Education, iv) Animal Science, v) Extension, vi) Horticulture, vii) Fisheries, and viii) Agricultural Engineering.
- A Deputy Director General (DDG) heads each Division, and they are entrusted with the overall responsibility for the preparation, scrutiny, review, and technical supervision and guidance of the research schemes and projects within their respective disciplines.

a) **Mission:** Sustainable growth of Indian agriculture by interfacing education, research and extension initiatives complimented with efficient and effective institutional, infrastructure and policy support that will create a proper fit between the humanity and its habitat.

b) **Vision:** ‘To harness science to ensure comprehensive and sustained physical, economic and ecological access to food and livelihood security to all Indians, through generation, assessment, refinement and adoption of appropriate technologies.’

c) Function or Objective of ICAR:

(i) To plan, undertake aid, promote and coordinate education, research and its application in agriculture,.

(ii) To act as a clearing house of research and general information relating to agriculture, animal husbandry, fisheries, home science and allied sciences through its publications and information system, and instituting and promoting transfer of technology programmes.

(iii) To provide, undertake and promote consultancy services in the fields of education, research, training and dissemination of information in agriculture.

(iv) To look into the problems relating to broader areas of rural development concerning agriculture, including post-harvest technology.

by developing cooperative programmes with other organizations such as the Indian Council of Social Science Research (ICSSR), Council of Scientific and Industrial Research (CSIR), Bhabha Atomic Research Center (BARC) and the Universities.

ICAR has following institutions

1. State Agriculture University (64)
2. Deemed Universities (04)
3. ICAR Institutions (64)
4. National Research Centres (15)
5. National Bureaux (06)
6. Directorates/Project Directorates (13)
7. Central Agricultural University (03)
8. State Agriculture University (64)

1. State Agriculture University (64)

SN	Name of University	State
1	Acharya NG Ranga Agricultural University, Guntur	Andhra Pradesh
2	Dr. YSRHU (APHU), Venkataramannagudem	
3	Sri Venkateswara Veterinary University, Tirupati	
4	Assam Agricultural University, Jorhat	
5	Bihar Agricultural University, Sabour, Bhagalpur	
6	Bihar Animal Sciences University, Patna	
7	Indira Gandhi Krishi Viswa Vidhyalaya, Raipur	
8	Chhattisgarh Kamdhenu Viswavidyalaya, Durg	
9	Sardar Krushinagar Dantiwada Agricultural University, Dantiwada	
10	Anand Agricultural University, Anand	
11	Navsari Agricultural University, Navsari	
12	Junagarh Agricultural University, Junagarh	
13	Kamdhenu University, Gandhinagar	
14	Chaudhary Charan Singh Haryana Agricultural University, Hisar	
15	Lala Lajpat Rai University of Veterinary & Animal Sciences, Hisar	
16	Haryana State University of Horticultural Sciences, Karnal	
17	Ch. Sarwan Kumar Himachal Pradesh Krishi Viswavidyalaya, Palampur	
18	Dr. Yaswant Singh Parmar University of Horticulture & Forestry, Solan	
19	Birsa Agricultural University, Ranchi	
20	Sher-e-Kashmir University of Agricultural Science & Technology, Srinagar	
21	Sher-e-Kashmir University of Agricultural Science & Technology, Jammu	
22	University of Agricultural Sciences, Bangalore	
23	Karnataka Veterinary, Animal and Fisheries Sciences University, Bidar	
24	University of Agricultural Sciences, Raichur	
25	University of Agricultural Sciences, Dharwad	
26	University of Horticulture Science, Bagalkot	

27	University of Agriculture & Horticulture Sciences, Shimoga	
28	Kerala Agricultural University, Thrissur	
29	Kerala University of Fisheries and Ocean Studies, Panangad, Kochi	
30	Kerala Veterinary and Animal Sciences University, Pookode, Wayanand, Kerala	
31	Rajmata Vijayaraje Scindia Krishi VishwaVidyalaya, Gwalior	
32	Nanaji Deshmukh Pashu ChikitsaVisvaVidyalaya, Jabalpur	
33	Jawaharlal Nehru Krishi Viswa Vidyalaya, Jabalpur	
34	Dr. Balaesahib Sawant Kokan KrishiVidyapeeth, Dapoli	
35	Maharashtra Animal & Fisheries. Sciences University, Nagpur	
36	Vasant Rao Naik Marathwada Krishi Vidyapeeth, Parbhani	
37	Matatam Phule Krishi Vidyapeeth, Rahuri	
38	Dr. Punjabrao Deshmukh KrishiViswaVidyalaya, Akola	
39	Orissa University of Agricultural & Technology, Bhubaneswar	
40	Guru Angad Dev Veterinary and Animal Sciences University, Ludhiana	
41	Punjab Agricultural University, Ludhiana	
42	Maharaja Pratap University of Agriculture & Technology, Udaipur	
43	Swami Keshwanand Rajasthan Agricultural University, Bikaner	
44	Rajasthan University of Veterinary & Animal Sciences, Bikaner	
45	SKN Agriculture University, Jobner	
46	Agriculture University, Kota	
47	Agriculture University, Jodhpur	
48	Tamil Nadu Agricultural University, Coimbatore	
49	Tamil Nadu Veterinary & Animal Sciences University, Chennai	
50	Tamil Nadu Fisheries University, Nagapattinam	
51	Sri Konda Laxman Telangana State Horticultural University, Hyderabad	
52	Sri PV Narsimha Rao Telangana Veterinary University, Hyderabad	
53	Professor Jayashankar Telangana State Agricultural University, Hyderabad	
54	G.B. Pant University of Agriculture & Technology, Pantnagar	
55	VCSG Uttarakhand University of Horticulture & Forestry, Bharsar	

56	Chandra Shekhar Azad University of Agricultural & Technology, Kanpur	
57	Narendra Deva University of Agriculture & Technology, Faizabad	
58	Sardar Vallabhbhai Patel University of Agriculture & Technology, Meerut	
59	U.P. Pt. Deen Dayal Upadhyaya Pashu Chikitsa Vigyan Vishwa Vidhyalaya Evam Go Anusandhan Sansthan, Mathura	
60	Banda University of Agricultural and Technology, Banda	
61	Sam Higginbottom University of Agriculture, Technology & Sciences, Allahabad	
62	Bidhan Chandra Krishi Viswa Vidhyalaya, Mohanpur	
63	West Bengal University of Animal & Fishery Sciences, Kolkata	
64	Uttar Banga Krishi Viswavidhyalaya, Cooch Behar	

Agriculture University In Maharashtra (04)

1. Mahatma Phule Krishi Vidyapeeth, Ahmednagar, Rahuri.
2. Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth, Dapoli. (18 May 1972).
3. Vasantao Naik Marathwada Krishi Vidyapeeth, Parbhani (1972)
4. Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola. (20 October 1969)

2. Deemed Universities (04)

- (i) Indian Agricultural Research Institute (IARI), New Delhi;
- (ii) Indian Veterinary Research Institute (IVRI), Izatnagar;
- (iii) National Dairy Research Institute (NDRI), Karnal; and
- (iv) Central Institute of Fisheries Education (CIFE), Mumbai.

2. ICAR INSTITUTES (64)

1. Central Agroforestry Research Institute , Jhansi
2. Central Arid Zone Research Institute, Jodhpur
3. Central Avian Research Institute, Izatnagar
4. Central Coastal Agricultural Research Institute, Ela, Old Goa, Goa
5. Central Inland Fisheries Research Institute, Barrackpore
6. Central Institute of Brackishwater Aquaculture, Chennai
7. Central Institute for Arid Horticulture, Bikaner
8. Central Institute for Research on Buffaloes, Hissar

9. Central Institute for Research on Cattle, Meerut, Uttar Pradesh
10. Central Institute for Research on Goats, Makhdoom
11. Central Institute for Women in Agriculture, Bhubaneswar
12. Central Institute of Agricultural Engineering, Bhopal
13. Central Institute of Cotton Research, Nagpur
14. Central Institute of Fisheries Technology, Cochin
15. Central Institute of Freshwater Aquaculture, Bhubneshwar
16. Central Institute of Research on Cotton Technology, Mumbai
17. Central Institute of Sub Tropical Horticulture, Lucknow
18. Central Institute of Temperate Horticulture, Srinagar
19. Central Institute on Post harvest Engineering and Technology, Ludhiana
20. Central Island Agricultural Research Institute , Port Blair
21. Central Marine Fisheries Research Institute, Kochi
22. Central Plantation Crops Research Institute, Kasargod
23. Central Potato Research Institute, Shimla
24. Central Research Institute for Jute and Allied Fibres, Barrackpore
25. Central Research Institute of Dryland Agriculture, Hyderabad
26. Central Sheep and Wool Research Institute, Avikanagar, Rajasthan
27. Central Soil Salinity Research Institute, Karnal
28. Central Tobacco Research Institute, Rajahmundry
29. Central Tuber Crops Research Institute, Trivandrum
30. ICAR Research Complex for Eastern Region, Patna
31. ICAR Research Complex for NEH Region, Barapani
32. Indian Agricultural Research Institute, Jharkhand
33. Indian Agricultural Statistics Research Institute, New Delhi
34. Indian Grassland and Fevener Research Institute, Jhansi
35. Indian Institute of Agricultural Biotechnology, Ranchi
36. Indian Institute of Farming Systems Research, Modipuram
37. Indian Institute of Horticultural Research, Bengaluru
38. Indian Institute of Maize Research, New Delhi
39. Indian Institute of Millets Research, Hyderabad
40. Indian Institute of Natural Resins and Gums, Ranchi
41. Indian Institute of Oil Palm Research, Eluru , Pedavegi, West Godawari
42. Indian Institute of Oilseeds Research, Hyderabad
43. Indian Institute of Pulses Research, Kanpur

44. Indian Institute of Rice Research, Hyderabad
45. Indian Institute of Seed Research, Mau
46. Indian Institute of Soil Sciences, Bhopal
47. Indian Institute of Soil and Water Conservation, Dehradun
48. Indian Institute of Spices Research, Calicut
49. Indian Institute of Sugarcane Research, Lucknow
50. Indian Institute of Vegetable Research, Varanasi
51. Indian Institute of Water Management, Bhubaneswar
52. Indian Institute of Wheat and Barley Research, Karnal
53. National Academy of Agricultural Research & Management, Hyderabad
54. National Institute of Abiotic Stress Management, Malegaon, Maharashtra
55. National Institute of Agricultural Economics and Policy Research, New Delhi
56. National Institute of Animal Nutrition and Physiology, Bengaluru
57. National Institute of Biotic Stresses Management, Raipur
58. National Institute of High Security Animal Diseases, Bhopal
59. National Institute of Research on Jute & Allied Fibre Technology, Kolkata
60. National Institute of Veterinary Epidemiology and Disease Informatics, Hebbal, Bengaluru
61. National Rice Research Institute, Cuttack
62. Sugarcane Breeding Institute, Coimbatore
63. Vivekananda Parvatiya Krishi Anusandhan Sansthan, Almora
64. Central Citrus Research Institute, Nagpur

3. National Research Centres (15)

1. National Centre for Integrated Pest Management, New Delhi
2. National Research Centre for Banana, Trichi
3. National Research Centre on Yak, West Khemang
4. National Research Centre for Grapes, Pune
5. National Research Centre for Litchi, Muzaffarpur
6. National Research Centre for Pomegranate, Solapur
7. National Research Centre on Camel, Bikaner
8. National Research Centre on Equines, Hisar
9. National Research Centre on Integrated Farming, Motihari
10. National Research Centre on Meat, Hyderabad
11. National Research Centre on Mithun, Medziphema, Nagaland
12. National Research Centre on Orchids, Pakyong, Sikkim

13. National Research Centre on Pig, Guwahati
14. National Research Centre on Plant Biotechnology, New Delhi
15. National Research Centre on Seed Spices, Ajmer

4. National Bureaus (06)

1. National Bureau of Agricultural Insect Resources, Bengaluru
2. National Bureau of Agriculturally Important Micro Organisms, Mau Nath Bhanjan, Uttar Pradesh
3. National Bureau of Animal Genetic Resources, Karnal, Haryana
4. National Bureau of Fish Genetic Resources, Lucknow, Uttar Pradesh
5. National Bureau of Plant Genetic Resources, New Delhi
6. National Bureau of Soil Survey & Land Use Planning, Nagpur, Maharashtra.

5. Directorates/Project Directorates (13)

1. Directorate of Cashew Research, Puttur
2. Directorate of Cold Water Fisheries Research, Bhimtal, Nainital
3. Directorate of Floricultural Research, Pune, Maharashtra
4. Directorate of Groundnut Research, Junagarh
5. Directorate of Knowledge Management in Agriculture (DKMA), New Delhi
6. Directorate of Medicinal and Aromatic Plants Research, Anand
7. Directorate of Mushroom Research, Solan
8. Directorate of Poultry Research, Hyderabad
9. Directorate of Rapeseed & Mustard Research, Bharatpur
10. Directorate of Soybean Research, Indore
11. Directorate of Weed Research, Jabalpur
12. Directorate on Onion and Garlic Research, Pune
13. Project Directorate on Foot & Mouth Disease, Mukteshwar

6. Central Agricultural Universities (03)

1. Central Agricultural University, Imphal, Manipur
2. Rani Laxmibai Central Agricultural University, Jhansi, UP
3. Dr. Rajendra Prasad Central Agricultural University, Pusa (Samastipur)

Lecture No 16

India Agriculture Concerns and Future Prospects

INDIAN AGRICULTURE CONCERNS:

1. Small and fragmented land-holdings
2. Seeds
3. Manures, Fertilizers and Biocides
4. Irrigation
5. Low Productivity
6. Subsistent in Character in Majority
7. Seed Replacement Rate
8. Traditional Bound
9. Lack of Diversification
10. Lack of Mechanisation
11. Regional variation
12. Farmers Education
13. Transportation & other infrastructure
14. Contribution in GDP Low but High percentage of employment rate
15. Lack of Marketing and Storage Facilities
16. Sustainability in Agriculture
17. Providing Loan in Wrong Way

1. Small and fragmented land-holdings:

India's total geographical area is 329 million hectares. Out of this, 195 million hectare is gross cropped area and 141 million hectare is net sown area. On the other hand, net irrigated area is only 65.3 million hectares

The average size of holdings was 2.28 hectares in 1970-71 which was reduced to 1.82 hectares in 1980-81 in 2016 average size of 1.15 hectares.

Total holdings, 85 per cent are in marginal and small farm categories of less than 2 hectares. The problem of small and fragmented holdings is more serious in densely populated and intensively cultivated states like Kerala, West Bengal, Bihar and eastern part of Uttar Pradesh where the average size of land holdings is less than one hectare and in certain parts it is less than even 0.5 hectare.

2. Seeds:

Seed is a critical and basic input for attaining higher crop yields and sustained growth in agricultural production.

Distribution of assured quality seed is as critical as the production of such seeds.

Unfortunately, good quality seeds are out of reach of the majority of farmers, especially small and marginal farmers mainly because of exorbitant prices of better seeds. In order to solve this problem, the Government of India established the National Seeds Corporation (NSC) in 1963 and the State Farmers Corporation of India (SFICI) in 1969.

Thirteen State Seed Corporations (SSCs) were also established to augment the supply of improved seeds to High Yielding Variety Programme (HYVP) was launched in 1966-67 as a major thrust plan to increase the production of food grains in the country.

3. Manures, Fertilizers and Biocides:

Indian soils have been used for growing crops over thousands of years without caring much for replenishing. This has led to depletion and exhaustion of soils resulting in their low productivity.

The average yields of almost all the crops are among the lowest in the world. This is a serious problem which can be solved by using more manures and fertilizers.

Manures and fertilizers play the same role in relation to soils as good food in relation to body. Just as a well-nourished body is capable of doing any good job, a well nourished soil is capable of giving good yields.

It has been estimated that about 70 per cent of growth in agricultural production can be attributed to increased fertilizer application. Thus increase in the consumption of fertilizers is a barometer of agricultural prosperity.

However, there are practical difficulties in providing sufficient manures and fertilizers all parts of a country of India's dimensions inhabited by poor peasants. Cow dung provides the best manure to the soils.

But its use as such is limited because much of cow dung is used as kitchen fuel in the shape of dung cakes.

Chemical fertilizers are costly and are often beyond the reach of the poor farmers.

4. Irrigation:

Although India is the second largest irrigated country of the world after China, only one-third of the cropped area is under irrigation.

Irrigation is the most important agricultural input in a tropical monsoon country like India where rainfall is uncertain, unreliable and erratic.

India cannot achieve sustained progress in agriculture unless and until more than half of the cropped area is brought under assured irrigation.

The success story of agricultural progress in Punjab area is under irrigation. Large tracts still await irrigation to boost the agricultural output.

5. Lack of mechanisation:

In spite of the large scale mechanisation of agriculture in some parts of the country, most of the agricultural operations in larger parts are carried on by human hand using simple and conventional tools and implements like wooden plough, sickle, etc.

Little or no use of machines is made in ploughing, sowing, irrigating, thinning and pruning, weeding, harvesting, threshing and transporting the crops. This is specially the case with small and marginal farmers.

It results in huge wastage of human labour and in low yields per capita labour force.

There is urgent need to mechanise the agricultural operations so that wastage of labour force is avoided and farming is made convenient and efficient.

Agricultural implements and machinery are a crucial input for efficient and timely agricultural operations, facilitating multiple cropping and thereby increasing production.

6. Soil erosion:

Large tracts of fertile land suffer from soil erosion by wind and water. This area must be properly treated and restored to its original fertility.

7. Agricultural Marketing:

Agricultural marketing still continues to be in a bad shape in rural India. In the absence of sound marketing facilities, the produce which is sold at throw-away price. In most cases, these farmers are forced, under socio-economic conditions, to carry on distress sale of their produce. In most of small villages, the farmers sell their produce to the money lender from whom they usually borrow money.

In the absence of an organised marketing structure, private traders and middlemen dominate the marketing and trading of agricultural produce.

The remuneration of the services provided by the middlemen increases the load on the consumer, although the producer does not derive similar benefit.

Many market surveys have revealed that middlemen take away about 48% of the price of rice, 52 % of the price of groundnuts and 60% of the price of potatoes offered by consumers.

According to an estimate 85 % of wheat and 75 % of oil seeds in Uttar Pradesh, 90% of Jute in West Bengal, 70 % of oilseeds and 35 % of cotton in Punjab is sold by farmers in the village itself. Such a situation arises due to the inability of the poor farmers to wait for long after harvesting their crops.

8. Inadequate storage facilities:

Storage facilities in the rural areas are either totally absent or grossly inadequate.

Under such conditions the farmers are compelled to sell their produce immediately after the harvest at the prevailing market prices which are bound to be low. Such distress sale deprives the farmers of their legitimate income.

The Parse Committee estimated the post-harvest losses at 9.3 % of which nearly 6.6 % occurred due to poor storage conditions alone. Scientific storage is, therefore, very essential to avoid losses and to benefit the farmers and the consumers alike.

At present there are number of agencies engaged in warehousing and storage activities. The Food Corporation of India (F.C.I.), the Central Warehousing Corporation (C.W.C.) and State Warehousing Corporation are among the principal agencies engaged in this task.

9. Inadequate transport:

One of the main handicaps with Indian agriculture is the lack of cheap and efficient means of transportation.

Even at present there are lakhs of villages which are not well connected with main roads or with market centres.

Most roads in the rural areas are Kutcha (bullock- cart roads) and become useless in the rainy season. Under these circumstances the farmers cannot carry their produce to the main market and are forced to sell it in the local market at low price.

10. Scarcity of capital:

Agriculture is an important industry and like all other industries it also requires capital.

The role of capital input is becoming more and more important with the advancement of farm technology.

Since the agriculturists capital is locked up in his lands and **stoc** to borrow money for stimulating the tempo of agricultural production.

The main suppliers of money to the fanner are the money-lenders, traders and commission agents who charge high rate to interest' and purchase the agricultural produce at very low price.

All India Rural Credit Survey Committee showed that in 1950-51 the share of money lenders stood at as high as 68.6 % of the total rural credit and in 1975-76 their share declined to%% of the credit needs of the farmers.

FUTURE PROSPECTS

To give stagnant agricultural growth a boost, a shift must be made from concentrating on the country's food security to focusing on the farmers income security.

1. Rainbow revolution holds the key (overall agricultural revolution)

In order to cross the declining productivity barrier there is a need to herald a rainbow revolution by making a shift from wheat-rice cycle to other cereals and pulses.

Since wheat and rice coupled with other crops water, fertiliser or power) regime, there is a huge incentive for the farmers in the irrigated region of Northwest India to grow these crops.

The policy response to this problem has always been to disincentive farmers from growing these crops by making meagre enhancements in the MSP.

Due to the rice milling industry in Haryana and Punjab, there is now proper established market in place for different varieties of rice that also incentives farmers to cultivate paddy.

Until such a marketplace is not created for other cereals and pulses, farmers are unlikely to make a shift to cereals and pulses.

2) Per drop more crop

The second major barrier is the scarcity of two major resources for agriculture, cultivable land and water. While the cultivable land per person is declining because of the fragmentation of farms due to rising population.

India also has much less per capita water as compared to other leading agrarian countries. This problem exacerbated because India has been exporting virtual water embedded in crops, which is marked by its feature of non-replenishment. Once it is exported, it cannot be recovered.

According to a report by Prashant Goswami and Shiv Narayan Nishad, in 2010, India exported about 25 cu km of water embedded in its agriculture exports, which is about 1 % of the available water every year.

Given this scenario, it is time to make a shift to micro irrigation so that the efficient and judicious use of scarce water resources can be made.

A study conducted by the National Mission for Sustainable Agriculture on micro irrigation 64 districts of 13.

Reveals that there were significant reductions in the use of water and fertiliser but the yield of crops increased up to 45% in wheat, 20% in gram and 40% in soybean.

However, high initial costs deter farmers to adopt this technology. big farmers can easily avail this technology, the government should consider giving subsidies to small farmers to boost the adoption of this technology.

3) Opening up of the markets

The National Agricultural Policy of 2000 stated that private sector participation will be promoted through contract farming and land leasing arrangements to allow accelerated technology transfer, capital inflow and assured market for crop production. However, there has not been any significant participation by the private sector in agriculture.

One of the major factors that deterred private players from entering the agricultural sector is the long pending reform of wholesale markets, which are regulated by the Agriculture Produce Management Committee (APMC) Act.

The APMC forces the farmers to sell their produce in government-controlled marketing yards. While the objective of APMC was to regulate markets and increase market yards, it has acted as a major obstacle to private investment.

Further, although the government has launched the National Agriculture Market, which provides farmers an electronic medium to sell their produce anywhere in India.

4) Research & Development is the future

One of the major barriers to boosting farm productivity is the lack of new technologies and major breakthroughs. While the National Agriculture Research System played a major role in the green revolution, in recent years there hasn't been any major breakthrough in research.

One of the main reasons for this is the lack of financial resources. If we compare the data of the percentage of agricultural GDP spending on research and development in Asia, then the figures are revealing.

While India spent 31 % of its agricultural GDP on research and development in 2010, in the same year China spent almost double than amount. Even our neighbour Bangladesh spent 38% of its agricultural GDP on research and development in that year.

As a result of this resource crunch there has not been diffusion of new agricultural innovations and practices that is critical for enhancing farm productivity.

Further, there is a lack of interest of students in pursuing research in agriculture. In the states where agriculture is relatively more important (as measured by their share of agriculture in state GDP), agriculture education is especially weak if measured by the number of students enrolled in agricultural universities.

5) Increasing produce per hectare

With a population that is expected to continue to rise till 2050, India needs to keep up its food production.

Unfortunately, the land area available to agriculture is not expected to increase.

Despite increasing food grain production over the years, statistics show that per capital food grain availability in India has been declining.

This brings us to the all important question of the need to increase productivity of agriculture and yields per hectare.

According to Food and Agricultural Organization (FAO), India's crop yield per hectare remains much lower than the global average. With around 2.4 tonnes of rice produce per hectare, India ranks much lower than even other BRICS nations in terms of productivity.

6) Greater penetration of crop protection measures

Despite being the fourth largest producer of agrochemicals globally, after the US, Japan and China, the usage of agrochemicals in India stands is actually very low, in fact, the one of the lowest in the world at just 0.58 kg per hectare against 4.5 kg per hectare in the US and 10.8 kg per hectare in Japan. The world's average consumption is 3 kg per hectare.

This shows there is a clearly large scope of growth in usage and demand. With limited availability of fertile land to cultivate food and feed an ever growing population, the only alternative we have is to increase productivity per hectare.

A study supported by industry body Assoc said that Indian farmers lose crop worth Rs 50,000 crore annually due to pest and disease infestation. The findings indicate a massive lack of awareness among farmers about crop protection measures, that if adopted can save huge crop losses.

We need to achieve greater penetration of usage of agrochemicals such as insecticides and pesticides to reduce crop losses and increase productivity per hectare.

7) Reducing dependence on monsoon

Making the agriculture sector less dependent on rainfall by diversifying sources of irrigation has been an area long discussed by experts.

To combat the persisting problem of lack of irrigation facilities to make up for the unpredictable rainfall, the government has introduced the decision to increase the area under irrigation by 28.5 lakh hectares and set aside Rs 20,000 crore for a new irrigation fund.

Additionally, it has committed to Rs 6000 crore for recharging the groundwater, with special emphasis on drought hit areas.

There have to be sustained efforts to improve groundwater levels through measures such as rainwater harvesting, discouraging deforestation, and developing artificial sources of irrigation.

8) Moving people out of agriculture

Reducing the rural population's dependence on agriculture has to be a prime development target for the government over the next 20 years.

This does not mean giving any less importance to agriculture; rather it means reducing under-employment on the farm and building alternative jobs outside the farm sector to wean away excess people to other sectors.

This can be done by strategically encouraging establishment of industries near rural areas that can provide viable employment, improving education in rural areas, providing vocational and skill training to the rural population and creating positive opportunities in nearby urban centres to encourage positive migration, as against distress migration.

SOME IMPORTANT SALIENT FEATURES OF INDIAN AGRICULTURE

(a) Subsistence Agriculture: As mentioned earlier, most parts of India have subsistence agriculture. This type of agriculture has been practised in India for several hundreds of years and still prevails in a larger part of India in spite of the large scale change in agricultural practices after independence.

(b) Pressure of population on Agriculture: Despite increase in urbanization and industrialization, about 70% of population is still directly or indirectly dependent on agriculture.

(c) Mechanization of farming: Green Revolution took place in India in the late sixties and early seventies. After more than forty years of Green Revolution and revolution in agricultural machinery and equipments, complete mechanization is still a distant dream

(d) Dependence upon monsoon: Since independence, there has been a rapid expansion of irrigation infrastructure. Despite the large scale expansion, only about one third of total cropped area is irrigated today. As a consequence, two third of cropped areas is still dependent upon monsoon. As you know, monsoon in India is

TYPES OF SOILS FOUND IN INDIA

Eight major types according to Indian Council of Agricultural Research (ICAR). These are Alluvial soils, Black soils, Red soils, Laterite soils, Desert soils, Mountain soils, Saline and Alkaline soils, Peaty and Marshy soils.

ALLUVIAL SOIL

These soils are formed by deposition of alluvium by rivers. It occupies over 15 Lakh square km area in India. Coarser in upper section and finest in delta. Soils are light to dark in colour, rich in Potash & Humus, poor in Phosphorous and Nitrogen. Soils are highly fertile, good for all crops (kharif & rabi). It contributes greatly in agricultural development. Soils are of Two types:

KHADAR & BHANGAR

- a) **KHADAR: Newer alluvium.** It has sandy and pale brown composition. These soils are found in lower areas and are flooded every year.
- b) **BHANGAR: Older alluvium.** These are Clayey & dark in colour. Coarse in nature. It contains Kankar (lime nodules), Pebbles and Gravels. These soils represent riverine alluvium of Himalayan rivers.

Areas: Punjab, Haryana, U.P, Bihar, W.B, Assam, Parts of Orissa, delta regions of S.India

Crops: Rice, Wheat, Sugarcane, Cotton, Jute.

BLACK SOIL

Also known as Regur or Black Cotton soil. These are dark grey to Black in colour having high clay content. It has highly moist retentive. Develops cracks in summer. It covers 5.4 lakh square km area. Soils are poor in Phosphorous, nitrogen and organic matter. However, rich in iron, lime, calcium, Magnesium, carbonates and alumina. These soils are highly suitable for cotton.

Areas: Deccan Trap which includes: Maharashtra, M.P, Parts of A.P, N. Karnataka, Parts of T.N and Rajasthan.

Crops: Cotton, Sugarcane, Groundnut, Millets, Rice, Wheat, Oilseeds.

RED SOIL

These are formed due to weathering of old crystalline rocks. Soils are more sandy and less clayey. These are poor in phosphorus, nitrogen and lime. However, rich in iron and small amount of Humus. It has slightly acidic and do not retain moisture. It covers 3.5 lakhs square km area. These are porous and friable.

Area : Tamil Nadu, Southern Karnataka, parts of Madhya Pradesh, Maharashtra, West Bengal, Eastern Rajasthan, North eastern States.

Crops : Ragi, Groundnut, millet, Tobacco, Potato, Rice, Wheat, Sugarcane.

LATERITE SOIL

It derived from Latin word meaning brick. Soils are formed under high temperature and rainfall with wet and dry spell. Silica is leached due to high rainfall. Remnants of iron and aluminum oxides left behind is known as Laterite. These are brown to yellowish colour. It becomes hard when exposed to atmosphere. These are used as building material. Soils are rich in Iron and poor in Lime, Potash, & Magnesium. It occupies 2.4 Lakh square km area.

Areas: Parts of Assam, Karnataka, T.N, A.P, M.P,Kerala.

Crops: Tea, Coffee, Cashew, Rubber and Coconut.

DESERT SOIL

These soils contains soluble salts. It was originated by Mechanical disintegration & wind deposit. Soils are Porous and coarse. It contains 90% sand and 5% clay. Soils are rich in Nitrates and Phosphates but poor in Nitrogen & Humus. These are friable, sandy and low moist content. It covers 1.4 Lakh square km area.

Areas: Arid and Semi arid regions of Rajasthan, Haryana, Punjab, Gujarat.

Crops: Drought resistant crops like millets and barley.

MOUNTAIN SOIL

These soils are found in hill slopes. These are formed by deposition of organic matter from forest. Soils are rich in humus. It has poor in Potash and Lime.

Areas: Assam, Kashmir, Sikkim and Arunachal Pradesh.

Crops: Tea, Coffee, Spices & Tropical Fruits.

SALINE and ALKALINE SOIL

These soils contains salts like Sodium, Magnesium, Calcium. It has sandy to loamy in texture. These are infertile, unfit for cultivation.

Areas: Parts of Gujarat, Rajasthan, Punjab, Haryana, U.P & Maharashtra.

PEATY AND MARSHY SOIL

Soils are occur in Humid region. These are formed by accumulation of organic matter. Soils are black in colour, highly acidic and heavy.

Areas: Kottayam & Alleppey in Kerala, Coastal Orissa, Sundarbans of W.B

THEORY NOTES

Course No. AGH 111 (1+0=1)

AGRICULTURAL HERITAGE

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